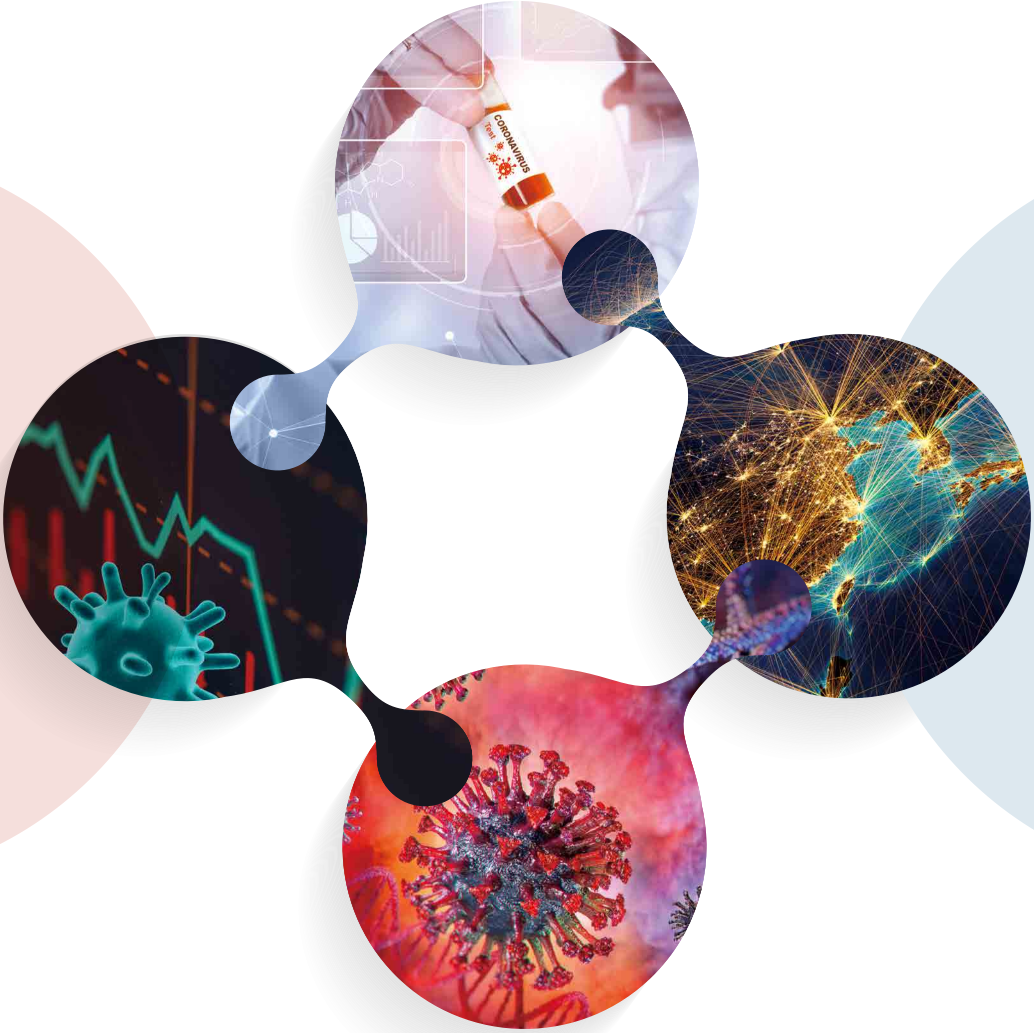
KOREA REPORT II

National Policy Transition in the COVID-19 Pandemic Era: **Responding to Grand Challenges**



경제·인문사회연구회
NRC National Research Council for Economics, Humanities and Social Sciences

STEPI Science and Technology Policy Institute

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"Global cooperation is not just the smart choice, but the right choice and it's the only choice we have."

(Tedros Adhanom Ghebreyesus (Director-General of the WHO), UN News, August 6, 2020)

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Publication Date	December 15, 2020
Publisher	Hwang Hee Cho
Publishing Office	Science and Technology Policy Institute (STEPI)
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Preface

It has been almost one year since the first COVID-19 case was confirmed in South Korea on January 20. Korea's coronavirus pandemic is still under progress and the situation has become more serious than ever before. As at 10th December 2020, the newly confirmed cases soared to nearly 700, and the death toll stood at more than 560 in South Korea. Securing vaccines developed by global pharmaceutical enterprises such as Pfizer and MODERNA that has passed all three phases of clinical trial is also a tough situation. Maybe, we are standing in the starting point of "Grand challenges" crossroad for national policy transition in the pandemic era. In order to explore national solution from the policy side, Korean Office of the Prime Minister and the affiliated NRC (National Research Council for Economics, Humanities, and Social sciences) planned for publication of policy research report in early 2020, which focuses on policy issues and countermeasures against infectious disease of COVID-19 and the relevant pandemic phenomena. To this end, one of the affiliated research institutes, STEPI (Science and Technology Policy Institute), has taken a leading role of contents creation for publishing the Korea Report. This report was devised to deal with national policy agenda regarding health and disinfection theme. The 1st version of Korea Report, under this background, was published on 26th May 2020, mainly highlighting the Korea's national responses to the COVID-19 crisis. Now STEPI intends to make public of 2nd version report named as "Korea Report II." The differentiating characteristics of Korea Report II are summarized as follow.

First, expanding scope of the subjects. Korea Report II makes publicized by dealing with more various COVID-19 issues from the perspective of national innovation activities. To this end, this report encompasses various spectrums of issues such as S&T policy and national innovation system transformation, policy measures and remedies for recovery of national economy during and after pandemic, international cooperation strategy in terms of S&T and ICT service, and methodological development to detect and monitor infectious disease patterns.

Second, setting up professional publication platform of the research. In the process of preparing for Korea Report II, project team suggested an idea to compose this report as a compilation type incorporating various studies on COVID-19. Accordingly, we collected information on celebrated scholars and practitioners globally in the innovation policy arena, thus contacted and had serious discussions with each prospective researcher. In consequence, three international researchers and one Korean researcher have approved to contribute to the report and succeeded to compose a journal type report.

Third, exploring ground breaking area for global issue tackling and entailed international cooperation. Some papers in this report applied national policy-level comparative studies and analyzed each nation's responses to thwart COVID-19 from various viewpoint: economic recovery, S&T and ICT, and research methods for empirical study. Through this approach, we could draw policy insights on globally urgent issues as well as have clues for international cooperation strategies and the related projects.

Korea Report II aims to help the global community's efforts to come up with effective countermeasures against this unprecedented pandemic by summarizing each nation's endeavors to tackle and response to COVID-19 and also identifying enabling factors behind. This report is composed of four chapters. Chapter 1 suggests a number of immediate STI policy considerations which is related to the South Korean context, based on policy reviews on emerging global STI trends. Chapter 2 identifies the impacts of COVID-19 and challenges posed to the business sector in the context of ASEAN countries with suggesting Korean policy direction. Chapter 3 suggests international cooperation direction based on the empirical analysis on the influential impact of ICT policy countermeasures and services on mitigating the dissemination of infectious disease. Finally, Chapter 4 provides new policy insights for national disinfection policy based on the analyzing real data of infection dissemination which was collected from Gyeonggi province in South Korea.

This report is written by a STEPI project team, which is led by Dr. Yongrae Cho, the Head of Future strategy Team. His project team includes Dr. Chansoo Park, the Director of Department of strategic planning, Myonghwa Lee, the Head of Office of national R&D research, and Ms. Aram Lee, a Researcher of Future strategy team. Without their dedicated efforts, this research report could not been published.

The project team sincerely appreciate all the expert authors that have contributed to this report by providing insightful manuscripts: Caroline Paunov & Sandra Planes-Satorra (Chapter 1), Patarapong Intarakumnerd & Upalat Korwatanasakul (Chapter 2), Mekuria Haile Teklemariam (Chapter 3), and Hyunjun Park, Ryunghyeok Im, & Yongrae Cho (Chapter 4).

Finally, project members specially thank to the publishers of Dr. Hwanghee Cho, the President of STEPI and Dr. Kyoungryung Seong, the Chairperson of NRC in South Korea. The Korea Report could be published thanks to their initial addressing necessity of this report, along with encouragement and support.

December 15, 2020

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KOREA REPORT II

Chapter 1

Science, Technology and Innovation Policy in Times of COVID-19: International Perspectives

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Abstract

As COVID-19 pandemic has become prevalent and threatened global health systems, the importance of urgency to find the solutions to these challenges has been continuously addressed and emphasized. Universities, scientists, and civil societies, accordingly, has made efforts to collaborate across many various disciplines. This shock resulting from the pandemic addresses and provides many challenges especially to science, technology and innovation (STI) ecosystem and the related platform. Under this background, this study explores the STI policy responses adopted across OECD countries from the start of the COVID-19 outbreak until September 2020, both to stimulate research and innovation to fight the pandemic, as well as to address the negative impacts of the crisis on STI actors. It also reflects on new policy trends that could shape the future of STI policy, and discusses a number of policy considerations of relevance for Korea. Based on the extensive analytical research on STI, this study suggests a number of immediate STI policy considerations which is relevant for the Korean context. We expect this study takes a role of the addition to recent issues on emerging STI policy trends that may shape longer-term policy directions in Korea as well as other countries. First, the need remains for policy to continue to support research and innovation to contribute solutions to the pandemic. Especially, Korea's engagement in international collaborations to share lessons learned on managing the pandemic and supporting solutions is important. Second, STI actors may face important challenges depending on the future evolution of the pandemic and the impacts of measures implemented across the globe to contain them. The policy interests should shed light on the inclusiveness challenges that would be led by COVID-19 shocks threatening the socially disadvantaged. Third, the responsiveness and agility of the STI system to this and future crises needs particular attention. In this regard, important tasks include improving the tools and data integrating system to monitor the real-time impacts to revisiting strategies and STI governance models

at national and global level. Those efforts will be necessary if STI in the post-crisis recovery is to promote the structural reforms needed for the transformation towards more sustainable, inclusive and resilient systems. The Korean New Deal goes in this direction by boosting investments in digital and green technologies. An assessment of the successes and failures of the Korean STI system in responding to the COVID-19 crisis can prove valuable for identifying where change is needed most.

I. Introduction

The COVID-19 pandemic became a global health threat by March 2020 and has, with extensive lockdown measures that were introduced in response in most countries, resulted in a global economic crisis of unprecedented nature. The urgency to find solutions to the COVID-19 health challenge, and the lockdown and other social distancing measures implemented to contain the spread of the virus from the start of the COVID-19 crisis to the present time of writing had important immediate effects on all actors in the science, technology and innovation (STI) ecosystem. Policies to support STI responses to the COVID-19 crisis have been implemented at unprecedented speed and scale.

Universities, public research institutions and pharmaceutical and biotech firms – sometimes in collaboration– have engaged in R&D efforts to rapidly develop vaccines, diagnostics and treatments for COVID-19. In early September 2020, there were 180 vaccine candidates (35 in clinical evaluation). More than 200 treatments and 500 clinical trials were under development already in April 2020, largely outnumbering those conducted during the year following the 2003 SARS and 2014 Ebola outbreaks.

Scientific efforts are also reflected in the surge of COVID-19-related research publications in the medical field and across many other disciplines. Many journals accelerated their peer-review process to ensure rapid dissemination of research results, and pre-prints - academic papers that have not been peer reviewed yet - became more common in the medical field (and beyond) in a matter of weeks. Moreover, many publications and research related to COVID-19 were made publicly available and provided in machine-readable searchable formats to allow for knowledge diffusion at scale.

Civil society actively engaged in these R&D and innovation efforts, for instance through online collaborative platforms that mobilised large communities of engineers and makers to build emergency medical supplies, or their participation in distributed computing projects, whereby they donate their spare computing power at home to run complex modelling for COVID-19 research in universities and research labs.

The shock resulting from the pandemic as well as lockdown and other social distancing measures affected entire STI ecosystems. Effects on research institutions and universities from lockdown measures ranged from interrupted research projects due to limited access to research labs, to cancellation of research mobility programmes, disruptions in human capital training and funding constraints due to expected reductions in income from student tuition fees. The pandemic also led to a re-orientation of health research to COVID-19 matters. Effects on innovative businesses in the early phase of the pandemic included financial constraints due to sharp reductions in demand, supply chain disruptions, and the slowdown in investor funding for (early-stage) start-ups. Yet many firms reacted rapidly to the new context by introducing - often digital - process and product innovations to maintain part of their activities or respond to new market demands. An acceleration in business uptake of digital technologies was also observed.

Impacts, however, differed significantly across STI actors. At the individual level, those in more vulnerable positions were often hit hardest by lockdown measures, including students from disadvantaged backgrounds, early-career and women researchers, and entrepreneurs. Firms were affected differently by the crisis depending on their sector of activity: low R&D intensive services sectors (in particular the tourism, entertainment and services requiring in-person interactions) and a number of manufacturing firms (e.g. aerospace) were highly disrupted, while digital services saw increases in demand for their products (e.g. digital collaboration tools, online shopping, telemedicine) and thus continued investing in innovation to improve their offering. Differences were also observed across regions, depending on their sectoral composition and the severity of local outbreaks of COVID-19 and subsequent restrictions implemented.

This short synthesis report explores the STI policy responses adopted across OECD countries from the start of the COVID-19 outbreak until September 2020, both to stimulate research and innovation to fight the pandemic, as well as to address the negative impacts of the crisis on STI actors. It also reflects on new policy trends that could shape the future of STI policy, and discusses a number of policy considerations of relevance for Korea.

More extensive analysis on STI in times of COVID-19 is provided in two policy papers on the impacts of the COVID-19 crisis on STI from January 2020 to September 2020 (Paunov and Planes-Satorra, 2020) and on the possible longer-term impacts of the crisis on STI and their policy implications (Paunov and Planes-Satorra, 2020). The analysis also leverages country policy information collected through the OECD Survey on STI Policy Responses to COVID-19 (“STIP Covid-19 Watch”).

This document is structured as follows. Section Ⅱ explores STI policies implemented across countries to stimulate research and innovation to provide solutions to the COVID-19 pandemic. Section III presents STI policies adopted to support STI systems withstand the COVID-19 crisis. Section IV discusses emerging STI policy trends. Section V provides a number of policy considerations for Korea.

II. STI Policies to Fight the COVID-19 Pandemic

1. Fast-tracking Research Funding Initiatives

Governments, firms and foundations have committed large amounts of funding for R&D activities aimed at developing vaccines, therapeutics and diagnostics for COVID-19. Several R&D funding trackers provide regularly updated estimates of the total amounts of funding allocated to COVID-19 R&D projects. According to the tracker developed by Policy Cures Research, a global health think tank, as of 18 September 2020 more than USD 9.1 billion had been committed by government, industry and philanthropic organisations to COVID-19 R&D projects. Nearly 60% of such funding has been allocated to vaccines R&D, and around half of the funds have come from organisations located in the United States (Policy Cures Research, 2020). As of 21 September 2020, data from the OECD Global Science Forum's database of COVID-19 research funding shows that total public and philanthropic investments in R&D projects amounted to USD 6.6 billion (OECD, 2020). A diversity of approaches have been adopted to allocate such funding.

Many governments have fast-tracked competitive research funding initiatives to support the development of COVID-19 vaccines, diagnostics and treatments. The challenge for accelerated procedures is to ensure fair, competitive and transparent merit review and selection processes. For instance, in March 2020, the French National Agency for Research (ANR) launched a Flash COVID-19 call of EUR 3 million (soon increased to EUR 14.5 million) allowing the evaluation, selection and funding of research proposals within a short period of time. Other examples are provided in 〈Table 1〉.

In some cases, support is channelled through existing funding mechanisms to accelerate responses. In Canada, for example, one of the measures of the 'Mobilize Industry' plan is the refocus of existing industrial and innovation programmes (e.g. Strategic Innovation Fund, Innovation Superclusters) to prioritize the fight against COVID-19. The US National Science Foundation (NSF) also fast-tracks research calls using their already existing Rapid Response Research (RAPID) funding mechanism.

Some government calls encourage existing grant holders to repurpose their research and innovation activities. For instance, the UKRI grants programme for ideas that address COVID-19 invites researchers holding existing UKRI standard grants to switch that funding to COVID-19 priority areas (UKRI, 2020).

To improve the visibility of research funding opportunities, some governments have invested in creating online platforms that list all relevant information on COVID-19-related STI activities, such as the European Commission's European Research Area (ERA) corona platform and Portugal's Science 4 COVID-19 portal (OECD, 2020).

Accelerating innovation

Table 1. **Fast-track R&D Funding Initiatives for COVID-19-related Research, Selected Examples**

Country	Programme	Funding agency	Focus	Amount of funding	Date launched
Spain	Special Spanish Research Programme on COVID-19	Institute of Health Carlos III – Ministry of Science & Innovation	Vaccines, diagnostics, treatments, social impacts, public health	USD 28.2 million (EUR 24 million)	19 March
United Kingdom	COVID-19 Rapid Response Calls	National Institute for Health Research and UKRI	Vaccines, treatments, diagnostics, social science research and other	USD 25.8 million (£ 20 million)	4 February
United States	Rapid Acceleration of Diagnostics (RADX)	National Institutes of Health (NIH)	Diagnostics	USD 248.7 million (as of 31 July 2020)	29 April

Note: * The French call focuses on the following four priorities: epidemiological and translational studies; the pathophysiology of the disease; infection prevention and control measures in the healthcare setting and in community settings; Ethics and the humanities and social sciences associated with the response.
Source: Paunov and Planes-Satorra (2020[1])

Most countries also implemented measures to stimulate quick innovative responses to the wide range of challenges posed by COVID-19 – from preventing the virus transmission, to producing essential supplies, addressing misinformation and handling effects of the lockdown. Approaches included the following.

Fast-track open competitions were launched to stimulate out-of-the box thinking and gather inputs from all parts of STI systems – from firms, to research teams and individual inventors. In some cases the challenge was clearly identified. For instance, the UK launched a call for innovative sanitising technologies allowing to clean ambulances rapidly after a patient with suspected COVID-19 has been transported. In Canada, the COVID-19 Challenge Programme posts specific challenges seeking near-to-market solutions from firms with less than 500 employees (e.g. low cost sensor system for COVID-19 patient monitoring) (Government of Canada, 2020). In Italy, under the “Innovate for Italy” initiative, a fast call competition was launched to identify best digital solutions available for telemedicine and monitoring applications for patients (MID, 2020). In other cases, such as Ireland’s COVID-19 Rapid Response

Call and UK's fast-track competition for business-led innovation in response to global disruption, applicants were asked to demonstrate the relevance of the challenge they addressed with their innovations.

Virtual hackathons were much used in the early phase of COVID-19 crisis. Organised by governments, non-profits universities and supra-national and international organisations, their objective was to source ideas from diverse contributors. Hackathons are typically 24- to 48-hour events in which participants are provided with data with which they have to create an innovative product. Winners are often compensated with funding to develop and scale their ideas. For instance, the European Commission organised the "EUvsVirus Hackathon" on 24-26 April 2020 to address around 20 COVID-19-related challenges. A total of 117 innovative solutions (out of more than 2,100 submitted) were selected as winners.

This European ("EUvsVirus") hackathon was followed by an online "Matchathon" in May 2020 to help winning teams match with corporates, investors and accelerators around the world to put their innovative solutions into production. This matching exercise sparked the development of 2,235 partnerships between the 117 winning teams and 458 supporting partners from the private and public sectors (European Commission, 2020). Some of the solutions are already publicly available, such as #WeStudyTogether - an online peer-to-peer learning community platform enabling educational institutions and teachers to engage their students remotely preventing student knowledge gaps and increasing retention.

Collaborations for research and innovation were also promoted. In Canada, the Pandemic Response Challenge program aimed to mobilise Canadian and international researchers from universities, business and government to work together to address specific COVID-19 challenges identified by Canada's health experts (Government of Canada, 2020). In the Czech Republic, a COVID-19 innovation vouchers call was launched to promote the sharing of knowledge and know-how between businesses and the research community to fight the virus and mitigate its impacts.

A number of calls focused explicitly on promoting international collaborations. For instance, in May 2020, the National Research Foundation (NRF) of Korea launched a Rapid Call for International Joint Research against COVID-19, to conduct epidemiological research involving Korean researchers collaborating with other researchers abroad. Another example is the Nordic Heath Data Research Projects on COVID-19, a call to foster research cooperation and health data sharing across Sweden, Finland, Denmark, Norway, Iceland, Estonia and Latvia.

Data and knowledge sharing was also encouraged. The EU and several partners established a COVID-19 data portal to enable rapid and open sharing of research data. In April 2020, the European Data Protection Board (EDPB) adopted guidelines on the processing of health data for research purposes in the context of the COVID-19 pandemic, addressing legal questions concerning international data transfers and the implementation of adequate safeguards (EDPB, 2020). National initiatives have also been adopted to encourage knowledge sharing across actors in the system. In Germany, the Federal Ministry of Education and Research (BMBF) allocated EUR 150 million to establish a National Research Network of University Medicine on COVID-19 – a centralised infrastructure to collect treatment data of patients with COVID-19 in a standardized way, and evaluate the action plans, diagnostic and treatment strategies of all German university hospitals.

Regulatory flexibilities were introduced where feasible to ensure rapid responses while maintaining safeguards. These leveraged existing emergency procedures of regulatory bodies, and were used to accelerate clinical trials to test new treatments and vaccines, the development of new products (e.g. diagnostics tests), and the production of products to address supply shortages (e.g. of medical equipment, medicines). Such mechanisms need to ensure compliance with ethical and scientific quality standards to safeguard the rights, safety and wellbeing of trial participants, patients and consumers. For instance, in Australia, the Therapeutics Goods Administration (TGA) has fast-tracked regulatory assessment of applications for COVID-19-related therapeutic goods, and set up exemptions for complying with normal regulatory processes and approvals to facilitate faster supply of tests, personal protective equipment and medical devices. In the UK, the Medicines and Healthcare Products Regulatory Agency published a package of regulatory flexibilities on 1 April to support the healthcare response to COVID-19, including through rapid reviews of clinical trials applications and expedited clinical investigations of medical devices.

Initiatives to facilitate access to research infrastructures, such as laboratories, databases and tools, were launched to help researchers accelerate their activities. The public-private COVID-19 High Performance Computing (HPC) Consortium in the US provides COVID-19 researchers worldwide with access to HPC, and the European Research Infrastructure on Highly Pathogenic Agents offers access to in-vitro and in-vivo research capacities to those conducting studies on COVID-19. CeADAR, Ireland's National Centre for Applied Data Analytics and Machine Intelligence, offered its AI expertise to help companies, government agencies and medical centres apply AI tools to help track the virus. In September 2020, three MedTech Open Innovation Testbeds (TBMED, MDOT, SAFE-N-MEDTECH) launched a call open to innovative companies and applied research institutions working to develop solutions to the pandemic. Selected innovators will have access to services to help them reach the market faster (MDOT, 2020).

III. Policies to Support STI Systems Withstand the COVID-19 Crisis

1. Supporting Public Research and Innovative Businesses in Times of Crisis

Aside from STI policy action to support research and innovation to respond to COVID-19 challenges, the immediate STI policy response has focused on keeping innovative businesses afloat and supporting researchers and public research organisation to quickly adapt to the new context. These are often part of broader stimulus packages aimed to boost the economy, such as the Coronavirus Aid, Relief, ad Economic Security (CARES) Act in the United States, and that directly or indirectly also support STI actors. The scale and speed of fiscal support provided by many countries is exceptional when compared to that provided during the 2008-09 financial crisis (IMF, 2020). Immediate policy measures to address negative impacts of Covid-19 on STI included those mentioned in the following paragraphs.

Flexibilities for existing beneficiaries of research and innovation programmes were rapidly introduced in most countries, and application deadlines were postponed. The Covid-19 pandemic affected the ability of existing recipients of R&D loans or grants to deliver their results on time, and of those preparing proposals to have them ready by the deadlines. In response, the EU postponed application deadlines for most calls to give more time to applicants to prepare their proposals. The Research Council of Norway also released a range of measures and principles to assist grant holders and applicants to new projects to deal with challenges due to the Covid-19 pandemic (e.g. projects that experienced delays, could not be carried out as planned, or needed to be put on hold). Similar flexibilities were introduced by most research funding bodies, such as the Australian Research Council and the German Research Foundation (Matthews, 2020). In the Netherlands, the repayment of Innovation Credits provided by the Dutch Enterprise Agency to innovative SMEs could be delayed for six months.

Support was provided to help higher education institutions and researchers cope with short-term challenges. These included helping HEIs provide tools and training to academic staff to be able to effectively deliver their teaching activities online (e.g. by making the best use of tools available, implementing alternative interaction and assessment methods), and to strengthen researchers' digital skills (e.g. to use online collaborative platforms). Immediate measures were also implemented to reassure students (from undergraduates to PhD students) and researchers (in particular early career researchers with fixed-term or project-based contracts) about the continuity of their programmes and/or funding. For instance, UK Research and Innovation provided grant extensions of up to six months for UKRI-funded PhD students in their final year, whose studies had been disrupted by the Covid-19 pandemic. The German Federal Ministry of Research and Innovation provided an additional EUR 100 million were made available for local student emergency funds to help students facing acute hardship. Many universities with well-established online programmes also made their training materials freely available, leveraging the advantages of scaling up online education compared to its offline alternative.

Measures were also adopted to protect research jobs and research projects impacted by the pandemic. For instance, in May 2020 Canada announced CAD 450 million (USD 341.6 million) in funding delivered as block grants for universities and health research institutions to retain research staff, keep essential research activities running during the pandemic, and help institutions ramp research back once physical distance measures are lifted. In view of expected income losses caused by a decline in international students, the UK launched a £280 million (USD 360.7) scheme that provides low interest loans to universities to support researchers' salaries and other costs such as laboratory equipment and fieldwork, as well as direct funding to maintain R&D projects.

Regarding business support, initiatives were introduced in many countries to facilitate access to funding to entrepreneurs and innovative firms to mitigate their liquidity problems. Such support can take different forms, such as loans, grants and repayable advances. For instance, in late March 2020 France launched an Emergency Start-up Relief Plan of EUR 4 billion (USD 4.75 billion), which includes the provision of state-guaranteed cash-flow loans, cash advances through the fast-tracked repayment of corporate tax claims that are refundable in 2020 (incl. the 2019 R&D tax credit), and early payments of the PIA (Investments for the Future Programme) Innovation Grants. In April 2020, the UK launched a £1.25 billion (USD 1.6 billion) package to support innovative firms hit by the pandemic. This includes a £500 million investment fund for high-growth companies, made up of funding from government and the private sector, as well as £750 million of grants and loans for SMEs focusing on R&D (GOV.UK, 2020). Germany also launched a EUR 2 billion (USD 2.4 billion) package to expand venture capital financing to support start-ups during the crisis (BMWi, 2020), and Israel launched a NIS 2 billion (USD 580 million) Rescue Plan for the High-Tech Industry for. The plan has allowed increasing loans granted by the Innovation Authority, and to offer large-scale debt financing to high-tech companies with significant assets (IP, funding, R&D) facing cash flow difficulties. Most countries provide additional support to SMEs more generally, explored in detail in OECD (2020).

Support for businesses –particularly SMEs and start-ups– to adapt to the Covid-19 context was also provided in some countries, to help mitigate the short-term negative impacts of the crisis. These could include supporting the use of online selling tools or adjusting production facilities to respond to new market demands. For instance, Enterprise Ireland provides Lean Business Continuity Vouchers of up to EUR 2,500 (USD 3,200) for companies to acquire training or advisory services support related to the continued operation of their businesses during the pandemic. It also offers Business Process Improvement Grants, which includes support to strengthen businesses' use of the internet as an effective channel for business development.

2. Towards More Agile and Responsive STI Policy

New approaches were also implemented to improve the STI policy responses to the pandemic. This included, first, a range of tools to monitor the real-time impacts of the crisis on different STI actors. For instance, Israel conducted monthly surveys and organised roundtables with main stakeholders to have a comprehensive picture of the main challenges faced by innovative businesses and how these evolved over time. Pulse surveys (or rapid response surveys) with reduced number of very targeted questions were also used to collect near real-time snapshots of the impact of the crisis on different actors, and track the evolution of those impacts over time. The US Census Bureau launched the Small Business Pulse Survey to collect weekly information on the challenges faced by small businesses during the pandemic with high level of geographical and sectoral detail (US Census Bureau, 2020). These experimental surveys are short and quick to fill out, as they only request timely information to track developments. Responses are subsequently linked to statistical information about the respondents (e.g. location, firm size, and sector of activity) collected previously through traditional census or business surveys.

Open data sharing by such initiatives as COVID-19 Open Research Dataset (CORD-19) - a dataset that contained at the end of September 2020 over 200 000 machine-readable scholarly articles on COVID-19 and related coronaviruses, including over 100 000 with full-text – has not only supported scientific activities but also been used by policy to identify the nature of scientific contributions to COVID-19. Early analysis of such data have pointed to the drop in female research activities and the high reliance on existing networks for research collaborations.

Second, digital tools were used to design and implement research and innovation policy. Such tools helped make decisions quicker and more effective, based on stronger evidence. For instance, the US National Science Foundation developed a visualisation tool that clusters all NSF-funded research projects addressing the COVID-19 pandemic in groups of similar topics, based on the application of machine learning techniques to abstracts of project proposals. This reduces risks of duplication in grant awards, facilitates the identification of synergies across projects and makes it easier to have a complete picture of the research areas being funded and their relative importance (Columbia University, 2020). SciSight is another visualisation tool that allows exploring the fast-evolving literature network around COVID-19 posted in CORD-19, and easily identify which research groups are working in which directions and their connections with each other.

Finally, efforts have also been made to enhance the responsiveness capacity of the public administration to this and future crises, leveraging on existing capacities in the STI system. Contact tracing applications, which follow the movement of infected people, were implemented in many countries, including France, Germany, Israel and Korea, to contain the spread of the virus and inform policy responses. In Finland, the Fast Expert Teams initiative was set up in March 2020 to support ministries respond to Covid-19 related challenges. It consists of a rapidly growing voluntary pro bono online expert network that gathers experts from universities, private and public sector organizations and ministries, to solve complex emerging problems requiring specialised and complementary expertise. In March 2020, the Portuguese Foundation for Science and Technology (FCT) launched the AI 4 COVID19, a competition with a budget of EUR 3 million (USD 3.6 million) for R&D projects in the field of data science and AI that contribute to improving the response of public administration bodies to the impact of COVID-19 and future pandemics.

IV. Emerging STI Policy Trends

1. Towards a "Steering" Role for STI Policy

The COVID-19 crisis may influence the focus of STI policy, as resilience, environmental sustainability and inclusiveness goals gain in importance in policy agendas for the recovery. Pursuing these objectives increases the directionality of STI policy.

STI can contribute to enhancing systems resilience – which refers to "the ability to anticipate absorb, recover and adapt to unexpected shocks" (OECD, 2020) – by increasing anticipation capacities (i.e. prevention of and preparedness to future shocks) and systems' agility (i.e. the capacity to quickly adjust in the advent of a shock). Support for research and innovation in green technologies and practices could significantly increase, as economic recovery packages (such as the Korean New Deal) integrate greening principles. STI policies could also play a key role in fostering inclusiveness by removing barriers to the participation of individuals, firms, sectors and regions underrepresented in innovation activities. The Centre for Women in Science, Engineering and Technology (WISET) in Korea is an example (find more examples of inclusive innovation policies in Planes-Satorra and Paunov (2017).

2. Strengthening systems' resilience

STI can contribute to anticipate future crises by engaging in preventive action and improving preparedness to crises. Strategic foresight may become increasingly embedded in policy-making processes to help policy makers prepare to strategically respond to a wide range of challenges that may emerge in the future, and detect early signals of those developments to provide timely responses. Anticipatory intelligence or strategic foresight exercises are not new tools in the STI policy domain, but they have traditionally focused on forecasting emerging technology and research fields.

Strengthening technological capacities in STI systems also contributes to enhanced anticipation and preparedness for future shocks. An example is Korean "Post-COVID-19 Science and Technology Strategical Direction", which identifies high priority technologies for R&D funding based on the assessment of scientific and technological experts.

Policy actions to build STI systems that are more agile and responsive to unexpected shocks include the following:

- Enhancing capacities and skills, especially for those in more vulnerable positions. Technology advances, but also their successful widespread adoption and rapid adaptation in case of disruption, highly depend on the research base and the levels of skills and capacities available in the economy. Strengthening such capacities, and ensuring that all segments of society have opportunities to develop them in order to contribute to and benefit from innovation, is essential to enhance systems resilience to shocks.

- Supporting international connectedness of research and innovation systems. No country has the the research base and scientific expertise at the cutting edge of all the areas that are relevant to respond to a specific crisis – ranging from experts in epidemiology, to seismology and artificial intelligence. In addition, as experienced during the COVID-19 crisis, international collaboration can accelerate responses and avoid duplication of efforts, also learning from past experiences in other countries (e.g. the 2015 MERS crisis in Korea).
- Strengthening the preparedness and flexibility of the public sector to respond to future shocks, including research and innovation funding agencies. Emergency R&D funding allocations and other measures introduced during the COVID-19 crisis (e.g. fast-track open innovation calls, flexibilities to grant holders and new applicants, etc.) should be assessed, in order to identify good practices as well as weaknesses and main challenges faced (e.g. shortages in evaluation capacities due to heavy increments in numbers of proposals).
- Reviewing STI governance systems to ensure they are agile in implementing coordinated STI-related measures across national and sub-national levels of government (and their alignment with policies in other areas). Such review also regards the processes to designate the composition and responsibilities of crisis task forces or councils, information sharing mechanisms across levels of governments and other actors in the STI ecosystem, and science advice and communication processes.

3. Systems Transformation Approaches

The COVID-19 crisis – as well as the 2008-09 global financial crisis – highlights the need for new approaches to better understand the nature of global challenges and articulate appropriate policy responses that take into account the complexity and interconnectedness of systems. Systems approach consists in setting policies considering that they are addressing issues that are part of a complex system of systems (economic, social, political, environmental, etc.), where changes in one component may directly or indirectly shape impacts in other parts of the system (Hynes, Lees and Müller, 2020).

The systems transformation approach points to the importance of developing balanced and well-aligned policy mixes, as discrete policy interventions are unlikely to bring about systems change. Even when single instruments succeed, they may result in unintended consequences and shift problems elsewhere in the system. Alignment is needed not only among innovation policies but also with other policy domains (e.g. research, education, competition, tax). In the past, intergovernmental committees and platforms have been used in many countries to ensure coordination among multiple policy instruments.

V. STI Policy Considerations for Korea

At the time of writing (October 2020), a number of immediate STI policy considerations are relevant for the Korean context, in addition to those relating to emerging trends explored in section 3 and that may shape longer-term policy directions in the country.

First, the need remains for policy to continue to support research and innovation to contribute solutions to the pandemic. As of October 2020, South Korea's response to COVID-19 stands out internationally as it has shown early success in flattening the epidemic curve rapidly without implementing strict lockdown measures as those in most other OECD countries (Global Change Data Lab, 2020). The Korean success in detecting, containing and treating the pandemic reflects the agility of Korean STI system in providing solutions to unexpected shocks, leveraging lessons from past crises (most importantly the 2015 MERS crisis). The high infection rates registered across many countries in November 2020, however, point to the continued threat posed by the pandemic to all countries. Korea's engagement in international collaborations to share lessons learned on managing the pandemic and supporting solutions remain important.

Second, STI actors may face important challenges depending on the future evolution of the pandemic and the impacts of measures implemented across the globe to contain them. A global recession would also impact Korean exports, which account for a large share of the economy (OECD, 2020). Such a reduction in demand for R&D intensive export sectors could reduce resources available to invest in their R&D and innovation activities. It is specifically important to ensure the COVID-19 shock does not lead to inclusiveness challenges by affecting disproportionately low-income households, early-career researchers, women researchers, SMEs and entrepreneurs.

Third, the responsiveness and agility of the STI system to this and future crises needs particular attention. The crisis confers an opportunity to reform those parts of research systems that operate sub-optimally, such as the low participation of women in research activities or the insufficient translation of R&D efforts into market innovations. Important tasks include improving the tools to monitor the real-time impacts to revisiting strategies and STI governance models at national and global level. Combining different new (and old) data to map entire systems at granular levels and in real-time, for instance, would allow capturing system dependencies. Those efforts will be necessary if STI in the post-crisis recovery is to promote the structural reforms needed for the transformation towards more sustainable, inclusive and resilient systems. The Korean New Deal goes in this direction by boosting investments in digital and green technologies. An assessment of the successes and failures of the Korean STI system in responding to the COVID-19 crisis can prove valuable for identifying where change is needed most. Peer learning from the experiences of other countries in dealing with the same crisis can contribute concrete options for Korea.

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Chapter 2

Business Impacts and Government Supports on Firms' Upgrading and Innovation under COVID-19 in Southeast Asia

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Abstract

This paper attempts to identify the impacts of the coronavirus disease 2019 (COVID-19) pandemic and challenges posed to the business sector in the context of Southeast Asia. Through this national-level comparative study, we intend to draw industrial policy implications for Korean government in the process of overcoming COVID-19 pandemic and recovering from economic recession. Due to the crisis and subsequent containment measures, most enterprises in the region had to suspend their businesses either temporarily or permanently and, consequently, are suffering from the liquidity crisis. Even though most enterprises realised the importance of innovation and technological upgrading and promoting new businesses like e-commerce, financial technology (Fintech), and teleworking, not so many enterprises really made it into reality. In response to the COVID-19 pandemic, all countries have been heavily focusing only on the short-term policy issues, such as economic lifelines and short-term recovery, while the longer-term and structural issues have not been paid enough attention to. Without structural adjustment of business models and operations to the new normal, enterprises are vulnerable and less resilient to future risks. Hence, there is still large room for policymakers to develop and improve longer-term policy measures that potentially reform the structure of both the public and private sectors. Although half of countries in the region have started a few initiatives regarding technological upgrading and innovation, the depth and dimension of the policy measures are still limited. On the one hand, less developing countries, i.e. Cambodia, Lao PDR, Myanmar, the Philippines, and Vietnam, have not explicitly initiated policy measures related to innovation and technological upgrading. On the other hand, Brunei Darussalam, Indonesia, Malaysia, Singapore and Thailand, to some extent, utilise support measures to promote innovation and technology in both of the public and private sectors. In general, sporadic incentive programmes (rather than holistic measures), generic capacity building programmes (rather targeted innovation

capability development), and economy-wide (rather than industry or sector-specific) innovation policy measures are observed in most countries. Comparing with ASEAN counties' cases, this study accesses the fiscal, monetary, and economic policy in South Korea, focusing on whether the governmental programmes aim national long-term recovery or not, with concluding policy implication in terms of national innovation system.

I. Introduction

Since the first quarter of 2020, most countries around the globe have been hit hard by the ongoing pandemic of coronavirus disease 2019 (COVID-19) posing great challenges to global public health systems. As of 4 October 2020, 35.2 million cases have been confirmed worldwide, while the total number of casualties surpassed 1 million people. To contain the scope and scale of the disease, several countries resorted to a series of lockdowns and severe restrictions on public activities. Consequently, the pandemic has caused short-term fiscal shocks and long-term adverse shocks to the growth of global economy as it disrupted global value chains, discouraged domestic and international consumption and investment, and imposed uncertainties in financial and foreign exchange markets. The world is facing a great trade-off between health emergency and economic crisis from the social distancing policy and the global and national lockdown.

The economic impacts of the COVID-19 pandemic are expected to be substantial. Due to the initial direct impact of the lockdowns, several economies could experience a decline in the level of output of between one-fifth to one-quarter as well as a drop in consumer expenditure of one-third (OECD, 2020a) and face negative per capita income growth in 2020. Developed countries such as France, Germany, Italy, Japan, South Korea, and the United States (US) are predicted to be in a recession with gross domestic product (GDP) at -5%, -6.8%, -1.5%, -1.8%, and -2.8% respectively (EIU, 2020). The World Bank (2020a) projects at least 2% drop in global GDP, while the International Monetary Fund (IMF) expects the global GDP to fall by 4.9%. In addition, the Organisation for Economic Co-operation and Development (OECD) (2020b) is more pessimistic and predicts that the global GDP would fall by 6%-7.6%, depending on the possibility of a second pandemic wave in the fourth quarter. However, a modest rebound in the growth of global economy of 2.8% - 5.4% in 2021 is expected (IMF, 2020a; OECD, 2020b). International investment and trade are also anticipated to slow down. In its recent forecast, the United Nations Conference on Trade and Development (UNCTAD) (2020) indicates a clear fall in global foreign investment by up to 40% in 2020, potentially leading to a drop of 45%-50% next year. According to the World Trade Organization (WTO) (2020), the volume of world trade is estimated to decline by 3% and 18.5% in the first quarter and the second quarter of 2020, while the drop in global merchandise trade may reach 32% by the end of the year. In terms of global employment, 5.3 million – 24.7 million people are expected to lose their jobs due to the impact of COVID-19 (ILO, 2020). Equivalent to 195 million global full-time workers, a 6.7% fall in working hours further aggravates the global employment situation.

The pandemic triggered the drastic changes in consumption patterns (e.g. higher reliance on online and delivery services) and employment (e.g. working from home and virtual meeting), posing significant uncertainty to all economic actors. The risks are even more pronounced among consumers and producers who have no access or limited access to technology and innovation. Even though the underlying costs and benefits of a new normal economy are still unclear, one positive side of the pandemic is that it transforms the global economy and accelerates the progress of the Industry 4.0 in unprecedented ways. Unless cease to exist, the affected private sector is forced to adjust their business model as well as upgrade their existing processes, products, functions, and positions in value chains through the adoption of new technology and innovation. Against this backdrop, this study realises the importance of technology as well as relevant policies and therefore places the topic of firm's upgrading and innovation in the centre of analysis and discussion in the paper.

This paper does not cover health issues related to containing the spreading of the virus or attempts in developing test kits, drugs, vaccines and medical equipment. Instead it aims to identify the impacts of the COVID-19 pandemic and challenges posed to the business sector in the context of Southeast Asia. Economic forecasts of key economic indicators such as GDP, trade, investment, and investment are presented, while the discussion of the results of business surveys conducted in some of countries in the region is provided. Moreover, the paper explores the current policy responses to the pandemic shock as well as to the economic shocks, particularly those related to firms' upgrading and innovation. By comparing the existing challenges with the current policy responses, this paper finds that all countries in the region have been heavily focusing only on the short-term policy issues, such as economic lifelines and short-term recovery, while the longer-term and structural issues have not been paid enough attention to. Moreover, even though half of them have started a few initiatives regarding technological upgrading and innovation, the depth and dimension of the policy measures are still limited. Only sporadic incentive programmes (rather than holistic measures), general capacity building programmes (rather than targeted innovation capability development), and economy-wide (rather than industry or sector-specific) innovation policy measures are observed in most countries. We assume that similar issues can be found in the Korean policy cases, too. Under the unprecedented pandemic situation, abrupt global shock which was derived from the COVID-19 has tremendously impacted on national economy. Accordingly, government, with no doubt, had no choice except short-term recovery programmes of remedies only with limited viewpoint and myopia. In this context, we need to draw policy implications and future directions based on the national comparative study and assessment of each government policy.

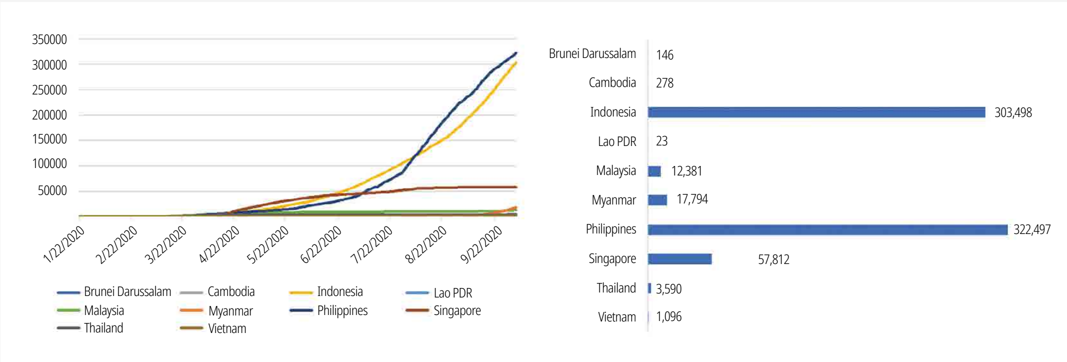
The paper is organised as follows. The next section briefly discusses the economic impact of the pandemic shock on the overall economy and the business sector in Southeast Asia. Furthermore, challenges emerged from and accelerated by the COVID-19 pandemic are also identified. Section III presents the current pandemic policy responses by each Southeast Asian country. Section IV discusses policy suggestions and concludes the paper.

II. Impact of COVID-19 and Challenges in Southeast Asia

1. Overall Impact

Despite the global large-scale COVID-19 confirmed cases and death toll, the situation of the pandemic seems under control in the Southeast Asian region, except Indonesia, Myanmar, and the Philippines [Figure 1]. Initially, countries in the region were criticised due to their late responses against COVID-19 before the disease quickly started to spread in the region. After March 2020, the number of confirmed cases has been rising in several countries, including Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Vietnam. With a time lag, the number of COVID-19 cases in Myanmar started to rise in August and soared up to 17,794 cases within a month, ranking fourth within the region. However, most countries have been able to prevent further expansion of the COVID-19 pandemic thanks to the effective lockdown and containment measures. The number of new cases is either zero or very low in many countries. For instance, even though Singapore was suffering from the high number of cases during the past few months, it has finally been able to flatten the COVID curve

Figure 1. COVID-19 pandemic curves and the number of confirmed cases (as of 4 October 2020)



Lao PDR = Lao People's Democratic Republic.
Note: The date format on the horizontal axis is month/date/year. COVID-19 = coronavirus disease;
Source: Authors, based on Dong, E., Du, H., & Gardner L. (2020) (accessed on 4 October 2020).

since the beginning of August. In contrast, Indonesia, Myanmar and the Philippines still have not reached the peak of their COVID curve; therefore, the countries are in urgent need of more effective policy responses.

While reducing the scale and scope of the pandemic as well as saving lives of citizens, the implementation of lockdown, containment, and social distancing measures come with great costs such as an economic crisis, a disruption of regional and global value chains, a collapse in consumption and confidence, among others. Economic forecasts issued in November 2019 depicted a moderate economic growth at 4.9% for the region in the period of 2020-24 (OECD, 2019). In May 2020, Asian Development Bank (ADB) released a policy brief on an updated assessment of the economic impact of COVID-19 and projected a 4.6% drop in the region's GDP in the case of shorter-containment scenario (where the outbreak lasts for 3 months), compared with a world without COVID-19, and a 7.2% fall in the case of longer-containment scenario (where the outbreak lasts for 6 months), with a double digit decline in some of the most hit countries, i.e. Cambodia (13.8%-20.2%), Malaysia (11.27%), the Philippines (11.52%),

Table 1. Impact of COVID-19 on GDP and GDP Growth of Southeast Asia, 2020

Country	Economy-wide impact (as % of GDP)		Forecast of GDP Growth
	Shorter-containment	Longer-containment	
Southeast Asia	-4.6	-7.2	na
Brunei Darussalam	-5.68	-8.45	na
Cambodia	-13.79	-20.19	-3.21
Indonesia	-5.77	-8.61	-1.74
Lao PDR	-5.94	-8.78	-2.15
Malaysia	-7.62	-11.27	-2.09
Philippines	-7.73	-11.52	-2.46
Singapore	-6.92	-10.26	-2.08
Thailand	-11.71	-17.22	-3.03
Vietnam	-5.96	-8.84	-2.69

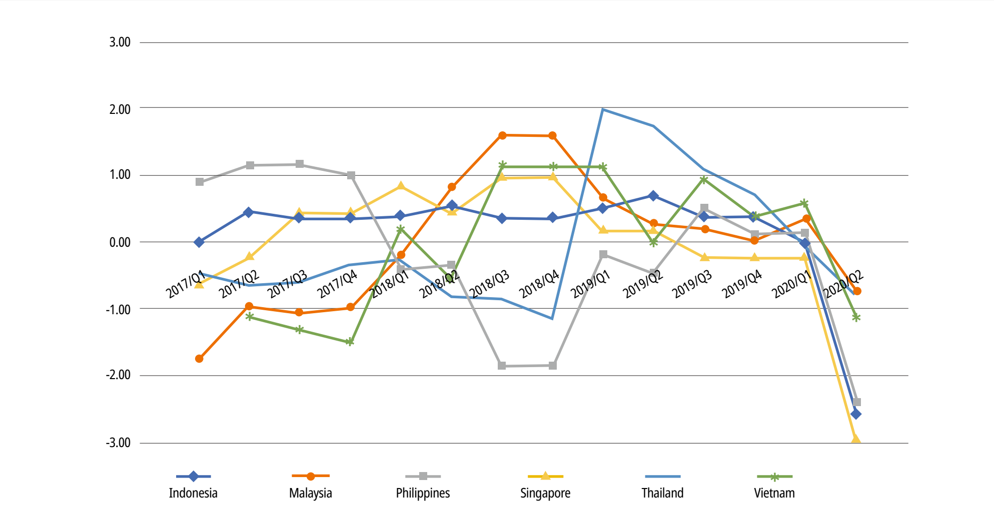
COVID-19 = coronavirus disease; na = not available; GDP = gross domestic product; Lao PDR = Lao People's Democratic Republic

Notes: The data of Myanmar is not available. The impact of COVID-19 on GDP and GDP growth is reported as the number compared with a world without COVID-19. A shorter-containment scenario represents a scenario where the outbreak lasts for 3 months, while a longer-containment scenario represents a scenario where the outbreak lasts for 6 months.

Source: Authors, based on Abiad et al. (2020), Park et al. (2020), and World Bank (2020a).

Singapore (10.26%), and Thailand (17.22%) 〈Table 1〉. Under the two scenarios, output in the region will be down by $163 billion to $253 billion (4.6% to 7.2% of the regional GDP). The initial direct impact of the shutdowns and quarantines largely came from the sharp decline in international tourism demand, followed by the drop of domestic demand and global spillovers. Even prior to the COVID-19 outbreak, with some recoveries, consumer sentiments across the region have been on their decline since the last quarter of 2018 or the first quarter of 2019. The pandemic led to further deterioration of consumer confidence with a sharp drop after the first quarter of 2020 [Figure 2]. Moreover, due to border closures, travel restrictions, and lockdowns, a significant impact of the outbreak is also observed on international trade. In Southeast Asia, a trade loss under the two scenarios is estimated to be approximately $229 billion to $344 billion 〈Table 2〉. Regarding the impact on employment, the pandemic will lead to employment losses equivalent to 11.6 million jobs to 18.4 million jobs under the shorter-containment and longer-containment scenarios respectively 〈Table 2〉. Additionally, wage incomes will drop between $25 billion to $39 billion across the region 〈Table 2〉.

Figure 2. Consumer Confidence in Southeast Asia (2017-2020)



Q = quarter
Note: Due to the difference of scale used, consumer confidence data is standardised.
Source: Authors' calculation, based on consumer confidence survey data from TheGlobalEconomy.com (2020).

Table 2. **Major Economic Impacts of COVID-19 on Southeast Asia under Two Scenarios**

Impact	Shorter-containment	Longer-containment
GDP ($ million), excluding policy measures	−163,223	−252,899
GDP (%)	-4.6	-7.2
Trade ($ million)	−229,495	−344,434
Trade (% of GDP)	−6.5	−9.7
Employment (million)	−11.6	−18.4
Wage Income ($ million)	−25,047	−38,986

COVID-19 = coronavirus disease, GDP = gross domestic product
Note: The estimates were released in May 2020.
Source: Authors, based on Abiad et al. (2020) and Park et al. (2020).

2. Impact on the Business Sector

Based on multiple firm-level surveys in each country, this section briefly discusses the current situation of the business sector and identifies its challenges and assistance needs under the COVID-19 pandemic. Even though the surveys' methodology of data collection such as sample size, timeframe, and scope of questions vary significantly, the survey results, to some extent, provide valuable information regarding the impact of the COVID-19 outbreak on enterprises in the region. To the best of our knowledge, enterprise surveys in Brunei Darussalam and Cambodia could not be identified.

Indonesia (ILO, 2020)[1]: Overall, two-thirds of enterprises reported temporarily or permanently closing their operations, with small enterprises being worst hit by COVID-19. Over a quarter of respondents reported a drop in revenue of more than 50% in the first quarter of the year, while 90% of firms experienced cash-flow shortages, creating a liquidity crisis and making access to finance and deferral of payment (e.g. utility bills, social security premium, taxes) among top priorities for government assistance. The significant drops in revenues and temporary business

1) Sustaining Competitive and Responsible Enterprises (SCORE) Programme Indonesia of the International Labour Organisation (ILO) and its partners conducted a survey of 571 enterprises in April 2020.

closure put pressure on the capacity of enterprises in retaining their employees. Approximately 63% of respondents have already let go their workers and some respondents will need to do so in the future. To response to the crisis, nearly one-third of companies highlighted online businesses as a key strategy, while one-fifth of companies resorted to product diversification to respond to emerging demand. However, the firms' capacity is still limited as more than two-thirds of enterprises cited teleworking or work-from-home (WFH) connectivity as a major challenge. Government assistance for survival and adaptation to "after-corona" are ranked the most needed support measures from the government.

Lao PDR (LNCCI, 2020)[2]: Firms in the tourism sector were worst hit by COVID-19, with 62% of respondents reporting high risk to cease business operation, as opposed to, on average, 45% in other sectors. Part of the uncertainty came from the reduction of revenue. 83% of enterprises reported lower sales due to COVID-19, with 27% facing sharp declines of more than 80% of their normal revenue in the first quarter of the year. The situation is particularly severe in the services sector where 37% of enterprises losing their revenue over 80%, as compared to the same period in 2019. Due to temporary shutdown, the most reported challenges were payment of salaries and loan and interest payment. Therefore, approximately 70% of firms is considering furloughing some employees, while almost 40% will need to reduce more than 50% of their employees. To resume their operations and continue operating after the pandemic, enterprises require government assistance in five areas, including tax relief package, rent and utility fee relief measures, soft loan and loan guarantees, grants and subsidies, and employment-related measures.

Malaysia (Department of Statistics Malaysia, 2020)[3]: As enterprises are facing constraints on cash and working capital, liquidity potentially becomes a serious

2) Lao National Chamber of Commerce and Industry (LNCCI) commissioned a survey of 474 enterprises in April 2020.

3) Department of Statistics Malaysia carried out an enterprise survey of over 4,094 responses in April-May 2020.

concern. Two-thirds of respondents reported no sales or revenue during the movement control order (MCO) period, while more than half expected to permanently shut down their businesses after 1-2 months of the MCO, given that they have to provide full-time or half pay leave to their employees. Constraints on additional credit were also alarming as almost 70% of enterprises use their own savings as the main source to accommodate operating cost and working capital, implying high credit demand in the future. In response to liquidity crisis, some enterprises (3.8%) have already terminated their employees' contract. Another 5.8% reduced working hours of their employees and 16.5% resorted to a mandatory unpaid leave. Salary payment posed the greatest challenge to enterprises (76.6%). Additionally, lack of customers and rental payment were the second and third most concerned issues, cited by about two-thirds of respondents. During the MCO period, firms' adaptation can be observed through operation adjustment, e.g. 12% of enterprises utilised an online platform to boost their sales and 33.5% allowed their employees to work from home. Relating to government relief measures, 83.1% of enterprises stated that financial assistance and subsidies was the most needed measure, followed by policy on taxation (67%), policy on loan programme (39.1%), and amendment of policies and related laws (30.6%).

Myanmar (World Bank, 2020b)[4]: According to the survey, 16% of businesses closed temporarily for an average of two months and anticipated that it will take at least a month to resume their operations. Moreover, the survey results revealed that over one third of firms anticipated to suspend business operations temporally or permanently within 1-3 months. Over 80% of respondents had experienced drop in sales, while almost half faced cash-flow shortages. Constraints on additional credit were also pronounced among local firms: around one third of firms experienced reduction in access to credit, and almost two thirds rely mainly on loans from friends and family. The agriculture sector was the most affected in terms of cash-flow shortages (64% of

4) The World Bank conducted a firm-level survey covering 500 firms from a wide range of industries and firm sizes during May-June 2020.

agriculture firms) and diminished access to credit (42% of agriculture firms). Overall, 69% of respondents reported over 55% of year-on-year drop of profits in the second quarter. The services sector had the highest share (71%) of companies reporting decline in profits compared to last year. To adapt to COVID-19, enterprises took several different measures. About 36% of enterprises tried to survive by starting or increasing delivery services. Moreover, around one out of five enterprises changed its production or services offered partially or completely to the change of consumer behaviour and emerging demand. Only few firms adopted online or digital platforms (19%) and WFH (6%). Regarding government support, almost half of businesses were not aware of support programmes for firms affected by COVID-19 and only 9% of firms applied for the programmes. Access to loans and credit guarantees were the most common policy support desired, cited by 38% of respondents. The second and third most common request were for tax deferral/deduction or relief (11%) and utility subsidies (9%).

The Philippines (ADB, 2020)[5]: About two-thirds of surveyed enterprises suspended business operations temporarily, while 29% are currently operating below their normal capacity, of which 78% operating at or below 50% of their capacity. Over 75% of respondents estimated a negative impact to their April sales (80.4% on average by value) compared with March. The temporary suspension resulted in a liquidity crisis for most of businesses (70%) as they did not have enough cash to remain operative. Difficulties in obtaining loans have exacerbated the situation as over half (57.3%) reported that it was more difficult to borrow than 2019. Moreover, the crisis forced enterprises to go through employment restructuring and redundancy by either furloughing their employees (41.9%), reducing working hours (41.4%), reducing salary and benefits (32.0%), or laying off (14.7%). Regarding firm adaptation to the COVID-19 situation, over 80% were not ready for WFH, particularly microenterprises.

5) The Asian Development Bank carried out an enterprise survey from 28 April to 15 May 2020, covering 2,481 enterprises.

Even though digitalisation and technological upgrading have not been observed much among Philippine enterprises, e.g. only 14% of the enterprises utilised online platform to sell their products, the business sector showed their interests in digital transactions (42.5%) and relevant skill training for workers (36%). Of the concerns facing by enterprises, the biggest reported challenges involved a payroll subsidy for workers (57.3%), deferment of payments to government (51.7%), and low-interest and subsidised loans (36%).

Singapore (EuroCham (Singapore), 2020)[6]: Almost one-third of respondents reported that they were significantly affected by the COVID-19, whereas 48% were moderately affected, particularly in terms of sales, cancelled activities, events, and meetings, among others. Overall, 88% of enterprises experienced reduction in revenue, with 18% facing declines of 30% and above of their normal revenue and 40% moderately experiencing higher operating costs. Corporate tax rebate and delay of planned goods and services tax increase were ranked the most preferred support measures from the government with 52% and 40% respectively. Moreover, extension of training grants and incentives for companies for upskilling and subsidies (28%) and relief for import export companies (22%) were the next preferred measures.

Thailand (UNIDO, 2020)[7]: The survey results revealed that one out of two surveyed small-size and domestic downstream enterprises expected to only survive for 1 to 3 months if the containment measures will be extended. Firms with small size and low technology were worst hit by COVID-19 pandemic and had very limited access to government supports (only 26%). Within the manufacturing sector, the automotive sector got affected the most, showing the lowest production since 1987. The two most

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6) European Chamber of Commerce (Singapore) commissioned a poll of 50 representatives of European organisations in Singapore in February 2020. Although the survey does not represent all enterprises in Singapore, the survey results could be considered lower-bound estimates of the current business environment in Singapore.

7) In collaboration with Ministry of Industry, Industrial Estate Authority of Thailand and Small and Medium Enterprise Institute under Federation of Thai Industries, United Nations Industrial Development Organization (UNIDO) carried out an online survey from 15 April to 15 May 2020, receiving a complete respond from 314 firms.

commonly reported impacts of COVID-19 were demand reduction and the shortage of inputs. Lower demand led to cash-flow shortages and reduced credit access; therefore, enterprises have been facing difficulties to make payments to their employees and social security as well as to repay loans to commercial banks. Overall, above 90% of firms experienced reduction in sales of more than half of their normal sales in the same period of the year. Despite the great concern on payments to employees, most of enterprises (above 80%) did not consider layoffs as the highest priority mitigation measure. Utilisation of advanced technology was the main strategy adopted as a countermeasure to the COVID-19 pandemic, particularly in terms of the shortage of workers. Interestingly, almost 90% of large firms resorted to advanced technology, while half of SMEs were capable of using similar technology. Tax-rate reductions or tax deferrals and a reduction of social contributions were the two most common policy support desired, cited by 49% and 35% of respondents respectively. the third most common request was for reduction of rent and utility costs (32%) followed by better loan terms (27%).

Vietnam (Hill, Baird, and Seetahul, 2020)[8]: According to the survey, almost half of enterprises, particularly those from the services sector, have suffered considerably from the COVID-19 pandemic and containment measures. Due to the containment measures, over 60% suspended their businesses or operated at partial capacity. The decline in customer demand posed the biggest challenge to most enterprises (84%), followed by the disruption of supply chain. Cash-flow shortages led to closing down part of the business operation (53%), work rotation (50%), WFH (47%), and layoff (16%). Remarkably, enterprises found emerging opportunities from new technology such as teleworking, electronic commerce (e-commerce), and digital platform. E-commerce and digitalisation were the most common adjustment mechanism adopted by enterprises in response to COVID-19 impacts, with more than one-fourth of firms

8) The survey funded by the Australian Government, collected information regarding the impact of the pandemic and the challenges faced by private sector employers from 38 enterprises during May and July 2020.

reporting adopting these measures. However, one in five firms experienced problems with internet connections when working from home.

Overall, most of enterprises in Southeast Asia, especially SMEs, have suffered from the crisis and containment measures and therefore suspended their operations temporarily or permanently. Among the firms remaining operating, two of three enterprises are operating at half capacity or less, while one of three enterprises are facing severe liquidity crisis and may have to close their businesses within three months. The sharp drops in demand and revenue were among the greatest concerns and the main cause of cash-flow or liquidity problems. In response to the COVID-19 pandemic, most of enterprises are aware of the benefits as well as importance of innovation and technological upgrading such as e-commerce, financial technology (fintech), and teleworking. Nevertheless, very limited number of enterprises are ready to adopt the technology. Short-term economic lifelines and recovery e.g. grant and subsidies, soft loans, tax relief package, deferment of debt, rent, and utility fee payments, and employee-related measures, were the most common policy support desired by the private sector. Additionally, enterprises also call for government assistance in terms of long-term or post COVID-19 business models and operations with emphasis on human resource development, innovation, digitalisation, and e-commerce. 〈Table 3〉 summarises policy issues under COVID-19.

Table 3. Policy Issues under COVID-19

Time frame	Policy issue
Short-term	Economic lifelines
	Short-term recovery
Long-term (post COVID-19)	Human resource development
	Business strategy and business plan development
	Innovation and upgrading (e.g. new process, new product, new function, and new sector)
	Digitalisation and e-commerce
	New market creationNew market creation
	Industry 4.0-related technologies (e.g. blockchain, artificial intelligence, robotisation, and 3D printing)

Source: Authors.

III. Policy Responses to COVID-19

1. General Policies Supporting the Business Sector

In response to the needs of the business sector, government in Southeast Asia have initiated a wide array of measures to support enterprises, particularly SMEs, to alleviate economic impact of the COVID-19 pandemic on businesses. 〈Table 4〉 captures the overview of policy instruments in several countries in Southeast Asia. In general, policy instruments initially applied only to certain industries or sectors affected the most by the pandemic, e.g. tourism, logistics, food, and medical supplies, among others. However, nearly all countries later expanded their policy coverage to all industries. Overall, to sustain enterprises' liquidity, current measures focus on short-term stimulus, including direct and indirect financial supports together with information and guidance services during the crisis. In terms of direct financial supports, all countries deploy grant and subsidies, while over half initiate loan guarantees (six countries) and soft loans (all countries except Brunei Darussalam). For instance, Brunei Darussalam provides grant worth BND20,000 (USD 14,700) for enterprises considering adopting e-commerce and delivery services. Unconditional cash transfers, including wage subsidies, are adopted in Cambodia for specific sectors (i.e. the garment and tourism sectors); Indonesia for financial incentive programmes and subsidies for loans and taxes; Malaysia for digitalisation (RM500 million or USD 120 million) as well as technological upgrading (RM100 million or USD 24 million) of SMEs; Singapore for short-term liquidity and long-term capacity building; Thailand for workers, farmers, and entrepreneurs; and Vietnam for enterprises with temporary closure and yearly revenues below VND100 million (USD 4300). Additionally, several countries introduce soft loans or direct lending to enterprises via public institutions, whereas some countries provide loan guarantees to ensure access to financial resources for enterprises, especially SMEs.

Similarly, countries in the region also commonly utilise indirect financial supports, including debt moratorium, tax relief package, employee-related policy instruments, and measures regarding rent and utility fees. All countries have been engaging with all kinds of indirect financial measures, except Indonesia and Myanmar in which policy measures either to reduce, defer, or exempt rent and utility payments have not been deployed 〈Table 4〉. Debt moratorium also includes loan restructuring and reduction or exemption of bank fees and charges. Tax relief package typically involves corporate income tax, value-added tax (VAT), customs and excise duties, and other government charges such as foreign worker fees. Government may resort to either tax reduction, tax deferral, tax exemption, or a mixture of the three methods. Due to temporary business closure and employment redundancy, all countries introduced employee-related measures, e.g. measures related to social security and pension as well as wage and income support for enterprises to secure necessary employment and for workers who are temporarily unemployed. Even though the introduction of indirect financial supports is different across countries, the measures are generally implemented economy-wide and last for at least six months to one year.

As most enterprises do not have sufficient capacity of crisis management, information and guidance services are considered ones of the effective and useful short-to-long-term measures that governments can offer during the pandemic. However, only three-fifths adopt these measures 〈Table 4〉. Indonesia, for instance, established call centres for SMEs and cooperatives affected by COVID-19 to advise businesses on how to adjust their operations during the crisis. Malaysia, meanwhile, introduced a recovery plan designed for all business categories to ensure the sustainability and rapid recovery of business operations. In the Philippines, business advisory assistance and services have been provided to affected SMEs under the Livelihood Seeding Program with a budget of PHP203 million (USD 4.2 Million). Singapore also developed a guide on business continuity planning for enterprises. Within the guide, comprehensive information on both managements of potential operational risks and sanitation were provided. Like Indonesia, Thailand set up a hotline for SMEs with queries about government support programmes, e.g. soft loans, tax deductions, rent and utility payments, among others. Vietnam emphasises more on domestic and foreign employees as the government facilitates enterprises to find alternative employee resources in case of a lack of foreign experts.

In sharp contrast to shorter-term stimulus measures, longer-term stimulus measures are still lacking and shallow in many cases. Long-term stimulus measures or structural measures possibly help enterprises to address short-term challenges, e.g. reduction in demand and changes in consumption patterns, while supporting enterprises to structurally adjust their business models and operations to the new normal and build resilience against future pandemics and other risks. The long-term stimulus measures include capacity building and training programmes, market-related measures, consultancy, and formalisation.

Among the long-term stimulus measures, capacity building and training programmes were the most common policy supports offered by governments in the region 〈Table 4〉. All countries except Lao PDR started to offer capacity building and training programmes to enterprises, particularly SMEs. Nevertheless, the programmes' focus and depth vary across countries. Brunei Darussalam, for example, offers over 300 free online courses on marketing and sales, finance and accounting, innovation, design, and data skills to enterprises. Similarly, the Philippines also offers free online courses, while implementing scholarship programmes amounting to PHP3 billion (USD 62 million) to finance upskilling and reskilling programmes. Out of USD 64 million, part of the budget was spent on skill training programmes in the garments and tourism industries in Cambodia. Indonesia and Vietnam provide stimulus funds to enterprises offering training programmes to their employees, while Thailand organised training programmes for affected workers and assisted local technology start-ups. Malaysia and Singapore, meanwhile, are among the most active countries that put forth automation, digitalisation, firm-level technological upgrading, and individual-level skill upgrading through various capacity building programmes in combination with grant and subsidy programmes. Digitalisation was indirectly promoted in Myanmar as government applications were moved to an online platform.

The second most common long-term stimulus measure is related to the development of existing market efficiency and promotion of new market access. Seven out of ten countries resort to these measures and intensively assist enterprises to access larger and new markets, including international markets, through promotion of digital solutions (Singapore), e-commerce platforms and virtual business matching and networking 〈Table 4〉. Additionally, governments also ensure the existing market efficiency, especially in terms of the supply of raw materials for manufacturing as well as the production and distribution of goods to meet the current domestic demand under the

Table 4. Policy Issues under COVID-19

Policy instruments	Brunei	Darussalam	Brunei Darussalam	Cambodia	Indonesia	Lao PDR	Malaysia	Myanmar	Philippines	Singapore	Thailand	Vietnam
Shorter-term stimulus measures	Direct financial support	Loan guarantees			○		○	○	△	△	△	
		Soft loans		△	○	○	○	○	△	○	○	○
		Grant and subsidies	○	○	○	△	○	△	△	○	○	○
	Indirect financial support	Debt moratorium	○	□	○	○	○	△	○	△	○	○
		Tax relief package	○	○	○	○	○	○	○	○	○	○
		Employee-related measures	○	○	○	○	○	△	△	○	○	○
		Rent and utility payments	○	□		○	○		○	○	○	○
Shorter to longer-term stimulus measures	Information and guidance				△		○		△	○	△	○
Longer-term stimulus measures	Capacity building and training programmes		○	□	△		○	△	○	○	○	△
	Market-related measures		○		△		△		○	○	△	□
	Formalisation			△						△		

ASEAN = Association of Southeast Asian Nations; Lao PDR = Lao People's Democratic Republic

Notes: ○ = Measures cover all enterprises, with emphasis on SMEs
△ = Measures cover only SMEs
□ = Measures cover all enterprises, without emphasis on SMEs

Source: Authors, based on IMF (2020b),OECD (2020c), OECD and ACCMSME (2020), UN ESCAP (2020).

crisis.

One of the issues posed to micro and small enterprises is the difficulty of receiving government supports and accessing financial resources as most of them are still concentrated in the informal sector. According to 〈Table 4〉, only one-third of countries in Southeast Asia utilise support measures to promote formalisation. Cambodia gives incentive stimulus package to informal SMEs on condition that they are formally registered, whereas Malaysia and Singapore explicitly include the informal sector in their recovery plans designed to simultaneously promote company formalisation.

2. Policies Related to Innovation and Technological Upgrading

In response to the new normal during and after the COVID-19 pandemic, innovation and technological upgrading, particularly in terms of digitalisation, are expected to empower individuals to adapt their working styles, including reskilling and upskilling, and to allow enterprises to adjust their operations in order to stay competitive, relevant, and resilient in the domestic and global markets. Thus, the pandemic is considered an accelerator of the process of digitalisation and upgrading in all economic sectors. Despite the importance of innovation and technological upgrading, policymakers in Southeast Asia demonstrate different levels of engagement in technological development process. On the one hand, less developing countries, i.e. Cambodia, Lao PDR, Myanmar, the Philippines, and Vietnam, have not explicitly initiated policy measures related to innovation and technological upgrading. This is mainly due to the lack of government capabilities in terms of finance and human capital. More urgent and pressing issues such as the short-run cashflow and liquidity crisis as well as insufficient ability to formulate and implement effective technology and innovation policies themselves may prevent the governments from issuing such policies. On the other hand, Brunei Darussalam, Indonesia, Malaysia, Singapore and Thailand, to some extent, utilise support measures to promote innovation and technological development in both of the public and private sectors.

Brunei Darussalam: The government mainly resorted to incentive programmes to promote digitisation in the financial sector and business enterprises. For instance, transfer fees and charges were waived for a period of six months for all customers using digital or online banking services. Moreover, the government provided co-matching grant of up to BND 20,000 (USD 14,700) for businesses planning to move their operations from offline to online or e-commerce. To facilitate the process of digitalisation, on 1 April 2020, the Authority and Info-communication Technology Industry together with Darussalam Enterprise launched Brunei Darussalam's first local online e-commerce directory, www.ekadaiBrunei.bn, while offering more than 300 free business-related online courses to improve technological competences of employees and business owners.

Indonesia: Similar to Brunei Darussalam, Indonesia also utilised incentive programmes to induce the business sector to adopt more digital technologies, including e-commerce and digital payment (fintech). The government created a demand for online products by funding a 25% discount for online goods, which, in turn, encouraged both consumers and producers to utilise e-commerce platforms. Additionally, e-commerce was further promoted among producers through a wide range of virtual business matching events organised by the government and relevant agencies.

Malaysia: Malaysia put serious efforts into the promotion of innovation and technological upgrading through several grants and loans such as the SME Digitalization Matching Grant (RM 100 million or USD 24 million), the SME Technology Transformation Fund (RM 500 million or USD 120 million), and the Smart Automation Grant (RM 100 million or USD 24 million). RM 300 million (USD 72 million) was allocated as SME Automation & Digitalisation Facility Grants for SMEs to upgrade their production. These grants allowed SMEs to invest in technologically advanced hardware and software, machinery, infrastructure, connectivity, cybersecurity, and system integrations to raise their productivity and efficiency. To be in line with higher technological advancement, the government helped workers and business owners cultivate their digital skills through short courses under the Human Resources Development Fund (RM 50 million or USD 12 million). Moreover, the government established the Malaysia Digital Economy Corporation (MDEC) to serve as a focal point to devise digital strategy and solutions for SMEs. Managing the grant worth RM 10 million (USD 2.4 million), MDEC's current mission is to promote e-commerce platforms. Additional RM 40 million (USD 96 million) were allocated to further develop e-commerce platforms that involve the traditional agricultural sector, which in turn potentially benefits from the market expansion of agriculture and food products.

Singapore: Several support programmes strengthening digitalisation and operational resilience have been in place in Singapore. Like Malaysia, Singapore emphasised on the promotion of innovation and technological upgrading, particularly, for SMEs and fintech firms. The Enterprise Singapore launched an E-Commerce Booster Package in April 2020 to support SMEs' business transformation efforts to utilise e-commerce service platforms such as Amazon, Lazada Singapore, Qoo10, and Shopee, while ensuring knowledge transfer between the platforms and SMEs. With the help of the grant, SMEs bear only 10% of qualifying costs and directly receive assistance from e-commerce service platforms on product launching, promotion activities, order fulfilment, and sales data analysis. Furthermore, SMEs Go Digital Programme offered government funding of up to 80% for pre-approved solutions, both foundational and advanced digital solutions, to enhance SMEs' digital transformation and capabilities. Through the Productivity Solutions Grant, the programme covered a wide array of costs including the development of the entire digital system such as basic hardware and software, digital infrastructure, Internet connectivity, cybersecurity, integrations between traditional production and digital technology. In May 2020, Singapore introduced Digital Resilience Bonus to endorse enterprises that plan to further digitalise their operations and work in collaborations with digital platform solution providers. Additionally, few more programmes, namely SG Together Enhancing

Enterprise Resilience (STEER) Programme, Enhanced Productivity Solutions Grant (PSG) for SMEs, and Enhanced Enterprise Development Grant (EDG), were implemented to strengthen enterprises in a wide range of areas, e.g. business growth, capacity and capability upgrading, digitalisation, innovation and productivity upgrading, and market access. Enterprises were also supported through the SGUnited Traineeships programme which mainly aims to bridge the gap between skills developed throughout the educational system and skills needed in the labour market and therefore boosts employability for new graduates.

Thailand: Thailand's policies regarding innovation and technological upgrading specifically under the COVID-19 pandemic are not well articulated in terms of actual implementable programmes, compared with those of Malaysia and Singapore. Specific policy measures and programmes for supporting innovation in enterprises under and after Covid-19 are still under the process of policy planning. So far, the government promoted digitalisation through a subsidy of 10GB free Internet and empowered local tech start-ups through training programmes. Nevertheless, since 2016, Thailand has been pursuing Industry 4.0. Upgrading existing industries and stimulating the emergence of new industries are one of objectives of Thailand 4.0. There are twelve targeted industries, namely, new-generation automobiles; smart electronics; affluent, medical and wellness tourism; agriculture and biotechnology; food; robotics for industry; logistics and aviation; biofuels and biochemicals; digital; medical services; education; and defense. To implement, Eastern Economic Corridor Project (EEC) covering three provinces in the Eastern part of Thailand (Chachoengsao, Chonburi, and Rayong) was initiated in 2016. Tax incentives for firms and individual specialists and researchers, matching grants, land ownership permission, and others are given for investment promotion in the targeted industries (Korwatanasakul, 2019; Lee, Wong, Intarakumnerd and Limapornvanich, 2019). Apart from the manufacturing and services sectors, the government also supported SMEs in the agricultural sector by connecting the industry with technology start-ups to promote smart urban farming.

IV. Conclusion and Policy Recommendations

Amid the coronavirus disease pandemic, enterprises around the globe are experiencing disruptions in global value chains as well as sharp declines in domestic and international consumption and investment. Southeast Asia's GDP is predicted to drop 4.6% - 7.2% during the lockdown period of 3-6 months, while a double-digit decline in GDP can be observed in half of the countries in the region. The pandemic also resulted in substantial deterioration of consumer sentiment, international trade, employment, and wage incomes. Due to the crisis and subsequent containment measures, most enterprises in the region had to suspend their businesses either temporarily or permanently and, consequently, are suffering from the liquidity crisis. Consumption patterns and therefore business operations changed drastically in response to the new normal in which technology and innovation, particularly digitalisation, seem to be the common solution to the current crisis. For instance, enterprises found it necessary to modify their business models and operations, while utilising technology and innovation to upgrade their existing processes, products, functions, and positions in value chains. Even though most enterprises realised the importance of innovation and technological upgrading such as e-commerce, fintech, and teleworking, not many enterprises are prompt to implement changes.

This study identifies policy issues from the enterprise surveys conducted within each country. The most common policy supports proposed by the business sector are short-term economic lifelines and recovery, such as grant and subsidies, soft loans, tax relief package, deferment of debt, rent, and utility fee payments, and employee-related measures. Moreover, long-term policy issues regarding post COVID-19 business operations were also among the greatest concerns of enterprises. Government supports in terms of human resource development, business strategy and business plan development, innovation and upgrading (e.g. new process, new product, new function, and new sector), digitalisation and e-commerce, new market creation, and Industry

4.0-related technologies (e.g. blockchain, artificial intelligence, robotisation, and 3D printing) are commonly discussed and can be identified from the surveys.

In terms of the overall policy responses to COVID-19, the current policy measures mainly focus on short-term stimulus, i.e. direct and indirect financial supports, covering all enterprises with emphasis on SMEs. Most countries resort to direct financial supports, including grant and subsidies (all countries), soft loans (nine countries), and loan guarantees (six countries). Similarly, indirect financial supports, i.e. debt moratorium (all countries), tax relief package (all countries), employee-related policy instruments (all countries), and measures regarding rent and utility fees (eight countries), are also implemented commonly among countries in Southeast Asia. Despite the significance of information and guidance services as the effective and useful short-to-long-term measures, only six out of ten governments in Southeast Asia adopted these measures. In contrast, longer-term stimulus measures are still lacking and shallow in many countries in which capacity building and training programmes are the most utilised policy measures. Policy measures regarding formalisation, systematic consultancy, and business development programme were barely observed during the period of crisis. Regarding the policies related to innovation and technological upgrading, less developing countries, including Cambodia, Lao PDR, Myanmar, the Philippines, and Vietnam, show insufficient level of engagement as no explicit policy measures in this area have been initiated. On the contrary, Brunei Darussalam, Indonesia, Malaysia, Singapore, and Thailand, to some extent, have explicitly adopted policy measures to promote innovation and technological upgrading among enterprises and enhance digital and technological skills among workers.

The similar phenomena could be found in Korean cases, too. Formerly, Korea Report II which was published in May 2020 has reviewed Korea's major countermeasures against the COVID-19 from four aspects such as "Prevention," "Detection," "Treatment," and "Recovery." (Lee et al., 2020) Out of these four factors, the recovery perspective addresses issues on how to overcome socio-economic aftershocks. Along with direct health-focused responses to the COVID-19 through aforementioned 3 remedies, various policy supports which are related to the "Recovery" have also been developed and prepared to recover national economy and society affected by the unprecedented shock. More specifically, "the Bank of Korea (BoK) has led the monetary policies to secure financial liquidity and to prepare for a potential global financial crisis. At the same time, the Ministry of Economy and Finance (MOEF) has led the fiscal policies for industrial recovery by drafting a revised supplementary budget." (Lee et al., 2020) Along with the declaration of special disaster areas to be able to provide bailouts and financial supports, an emergency disaster relief fund was provided for the whole nation.

The issue, however, is that most of the policy support tools should not be only limited to a temporary expedient with short-term boosting effects. In order to balance between short-term and long-term policy programmes in the progress of national recovery, it is time to settle down to a nation-level comparative study. Through this, we should draw future directions based on the national comparative study and assessment of each government policy. Furthermore, international policy cooperation strategies can be more concretely developed.

In conclusion, in response to the COVID-19 pandemic, all countries have been heavily focusing only on the short-term policy issues, such as economic lifelines and short-term recovery, while the longer-term and structural issues have not been paid enough attention to. Without structural adjustment of business models and operations to the new normal, enterprises are vulnerable and less resilient to future risks. Hence, there is still large room for policymakers to develop and improve longer-term policy measures that potentially reform the structure of both the public and private sectors. Even though half of countries in Southeast Asia have started a few initiatives regarding technological upgrading and innovation, the depth and dimension of the policy measures are still limited. Most countries except Singapore and Malaysia, for instance, do not have concrete policy measures on innovation and upgrading in response to the new normal under and after Covid-19. In addition, only sporadic incentive programmes (rather than holistic measures), capacity building programmes (rather targeted innovation capability development), and economy-wide (rather than industry or sector-specific) innovation policy measures are observed in most cases.

We propose a few policy recommendations.

Firstly, it is reasonable to have short-term policy measures to guarantee the survival of firms during the pandemic. However, it is also increasingly important to provide more resources and time to formulate long-term policies for the post Covid-19 which become new normal, as firms will be more digitalized and new business activities and business models will emerge.

Secondly, to help firms to seize new opportunities under the post Covid-19, it is necessary to for government to provide holistic measures to support innovation. Targeted financial incentives, especially in the form of grants, to encourage firms to develop and deepen their technological capabilities underpinning several types of innovations (new products, new processes, new business models) should be encouraged. Non-financial incentives like government-subsidised training programmes, digital infrastructure (like 5-G telecommunication networks), consultancy/guidance services, conducive regulation, and market-creation measures like government procurements for innovative products/services (i.e., providing ‘first market’ for innovation from the private sector) should be simultaneously initiated.

Last but not least, apart from economy-wide or generic innovation policy measures, it is essential to develop and implement industry-specific policy measures. Though impact of changes derived from Covid-19 encompass most, if not all, industries, the nature of business opportunities and concomitant innovation processes differ across industries. It is imperative for government to study how innovation happen or will happen in different industries (both traditional and emerging), and later devise appropriate policy measures to promote and facilitate innovation processes in those industries.

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Chapter 3

Developing ICT Policy for Fighting COVID-19 in Africa: Collaboration Opportunity from South Korea's Best Practices

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Abstract

This paper assumes that in order to prevent COVID-19 cases especially in highly densely populated urban areas, ICT policy that includes ICT usage skill, female social media penetration, e-government and online services with cyber security need to be developed in addition to the network infrastructure coverage improvement. This is because awareness creation about how COVID-19 spreads and getting services online using broadband instead of the face-to-face interaction is the best way to thwart coronavirus. In this study, a total of 46 African countries have been taken as sample of the analysis, Korea has been taken as benchmarking country and used the 2020 data. With pooled and interaction effect cross-sectional data models, this paper examines which variables out of 12 explanatory variables were effective in controlling the spread of COVID-19 in Africa. Further, by interacting online service index variable with the network infrastructure development, this paper identifies favorable differential slope (policy) effects of online service and telecom infrastructure dev't on COVID-19 infection prevention and Control in Africa. The paper concludes that, among the 12, seven variables that include digital skill, e-government service, female social media penetration level, and network with secured server and capacity of 4G have significant effects on coronavirus infection prevention. Whereas urbanization in Africa has facilitated for social closeness and favorable effects on the spread of the virus. Finally, to mitigate the crisis in Africa, the researcher proposes a set of policies focusing on ICT best practices of South Korea that can be augmented to African countries.

I. Introduction

Coronavirus disease 2019 (pandemic) caused by severe acute respiratory syndrome coronavirus 2 has rapidly created a catastrophe for the society (Wu et al., 2020). The report of World Health Organization (WHO) indicates that a total of 38,457,000 COVID-19 confirmed cases with fatality rate (CFR) of 2.84 percent (1,092,332 total deaths) from 215 countries and territories until 10th of October, 2020. The proportional distribution of cumulative cases and its percent based on the regions include: Europe 6,313,257 (16.42) and 233,686 (21.4); North America 9,712,964 (25.30) and 328,831 (30.12); Asia 12,054,448 (31.35) and 216,842 (19.85); Latin America 8,738,070 (22.72) and 273,328 (25.02); Africa 1,604,442 (4.17) and 38,681 (3.54); and Oceania 33,098 (0.86) and 949 (0.87) of total cases and total deaths respectively. As observed from the data, this global pandemic has been attributed to the facets of complexity, uncertainty, and difficulty for evaluation which requires concerted and sustained contributions from various disciplines and worldwide collaboration (Gümüsay & Haack, 2020) until empirical truth is going to be established about it.

Even though the COVID-19 proportional distribution of cumulative cases of Africa in percent (4.17) relative to the global cases are far below the percentage share of Africa (17.2) population from the world, it is time to reset and reform healthcare systems across the continent using mobile technology and broadband as the fundamental features to build resilience amid the pandemic since COVID-19 has highlighted the often-parlous state of Africa's healthcare infrastructure. As COVID-19 explodes on, it gives Africa multiple reasons to turn to mobile innovations in order to advance the continent's e-health services. The overwhelming of healthcare systems in developed economies highlights a catastrophe is boiling in Africa, while the resurgence of the pandemic across the world suggests that Africa should have preparedness for a long fight. Hence, this is a time to revamp the African healthcare delivery systems – to enable equitable and high-quality delivery of healthcare to households to combat the pandemic, and to reinvent healthcare systems not only for controlling the spread of COVID-19 but also to have resilience for the post-pandemic era of 'Unparalleled economic shock'. We can achieve this by complementing old systems with new digital innovations by scaling-up the best practices of South Korea as reported by the Republic of Korea.[1)]

In relation to the role of ICT in controlling the pandemic and to have resilient economic activities counter to the shock, we will highlight the four relevant themes that have evolved during the pandemic and the corresponding topics that are identified by Pan & Zhang (2020) which this study further adapted to the context. The issues addressed as an example include: developing ehealth digital platform for universal access that enables the COVID-19 cases recovery and sharing the accurate information; bridging the female social media penetration gap; developing digital skill; developing the egovernment and other online services, reliable and secured broadband network infrastructure development and managing cities appropriately with the COVID-19 more exposure risks. From each theme, it has been pointed out areas of collaboration opportunities in relation to ICT development between Africa and South Korea.

Finally, we propose the African countries to play proactive role by articulating their own demands and initiating the projects to make the goal practical to control the pandemic and tackle SDGs, which would in turn build capabilities in digital sustainability.

The structure of this study is as follows. Section 2 briefly reviews existing studies on the determinants of COVID-19 spreads focusing on ICT and discusses the new contributions of this research and its objectives. Section III introduces the research methodology including research hypotheses, models and data, Section IV presents

1) How Korea responded to a pandemic using ICT: Flattening the curve on COVID-19. http://overseas.mofa.go.kr/gr-en/brd/m_6940/view.do?seq=761548&srchFr=&amp%3BsrchTo=&amp%3BsrchWord=&amp%3BsrchTp=&amp%3Bmulti_itm_seq=0&amp%3Bitm_seq_1=0&amp%3Bitm_seq_2=0&amp%3Bcompany_cd=&a

the results of analyses, and Section V presents discussions and policy implications. Section VI concludes the paper.

II. Literature Review

The COVID-19 pandemic has changed many aspects of our day to day pursuits and the method we do them. The role of ICT sector fundamentally changed such as online services, information behavior, models of business doing, the role of cybersecurity, and data privacy (Davison, 2020). The potential opportunities for nations' ICT development is to use the demanding context to contribute to this ongoing COVID-19 pandemic challenge and try to build on the collective knowledge and institutional dynamic capacity that can respond not only this pandemic but also swiftly resilient to another similar national shocks (Davison, Hardin, Majchrzak, & Ravishankar, 2019). To catch up with such goals, African countries may need to initiate collaborative partnership with ICT industry best performing countries such as South Korea in tackling the coronavirus complex challenges through developing adaptive and interoperable applications and Information Systems. It could be suggested to have collaboration in two major areas of ICT that include: First, how we could exploit the potentials of ICTs for fighting against the common enemy of COVID-19, and second, how we could mainstream in the private and government services associated with the use of ICTs for emerging economic recovery and resilience strategies by addressing the following issues.

1. Developing Ehealthcare Services' Digital Platform

Developing ehealth digital platform to share accurate information about the pandemic and to aware the public about the importance of ensuring the physical distancing is one of the key strategies towards controlling the pandemic. With improving broadband coverage, African countries should be ready for quality healthcare systems powered by digital innovations especially using online health services delivery system. It is, therefore, the responsibility of both public and private players to seize this moment to mainstream ICT for the continent's healthcare system.

Even though there are a number of studies identifying the negative impacts of the COVID-19 virus and social distancing measures, there are no empirical data-based research on how ICTs may help to prevent the spread of the virus and connect the people through the internet as the alternative keeping the connections. Hence, this empirical based study investigates based on online services empirical based data on how ICTs may contribute to controlling the coronavirus case spreads.

2. Developing Interoperable Egovernment Services Ecosystems

In the case of the COVID-19 pandemic, a wide variety of distributed and decentralized entities are involved in the egovernment services delivery, data production and collection processes. Such service delivery national platform and data ecosystems are multi-layered and embedded in pluralistic institutional environments with diverse cultures, structures, and priorities (Gupta, Panagiotopoulos, & Bowen, 2020) which need to be interoperable to have effective system. As studied by Ting et al., (2020), there are several participating entities for such cases that include local and regional government authorities, technology developers and providers, healthcare institutions, private organizations, non-governmental organizations as well as local citizens. These entities, the data, and their interactions contribute to data ecosystems at the local, regional, and national levels which require national interoperable platform to integrate and synergize the effort for productivity improvement. In African, crises such as the COVID-19 pandemic, where enormous amounts of data are generated instantly from isolated sources, organization with authority and all the entities involved in the data ecosystems, should participate in the interoperable sustainable service delivery system development processes. Data collaborations between all the entities must be formed rapidly to effectively combat the pandemic (Hua & Shaw, 2020) in which South Korea have several best practices that could be scaled up to African countries.

3. Developing National Private and Public Online Services Delivery System

Individuals need to have digital basic skills and adapt to new information environments during the COVID-19 pandemic by using the emerging wide variety of online services technology. In such case, national and local government including nongovernmental institutions may need to give basic digital skill training for consumers that could enable them to perform their online information activities. As it is noted by the study of Pan, Cui, & Qian, (2020), it is very important to develop strategy on how individuals can adapt their information behavior in these new requirements of online service development for African countries citizens to survive the pandemic and prepare for the post- pandemic digital based sustainable development goals attainment.

Further, Caligiuri et al. (2020) stressed that the outbreak of the COVID-19 has forced many private sectors to launch new initiatives that enable them to accelerate the transformation to the digital workplace. One such initiative as an example is protecting the health of employees, keeping the business running, and preparing better for the recovery phase as organizations went through the rapid rollout of online working environment (Verbeemen & D'Amico, 2020). Hines (2020) highlighted that organizations have quickly implemented a variety of digital infrastructures and tools to maintain uninterrupted service to their customers.

However, Verbeemen & D'Amico (2020) reminded that a rapid switch to a remote working environment may raise many challenges as the implementation of online working in the pandemic is broader and deeper than most organizations realize.

Hence, such demand may need to have collaboration that may extract plentiful opportunities for ICT mainstreaming. The large-scale impacts of the COVID-19 pandemic will provide plausible and live opportunities where to start the initiatives with more diverse contexts to explore the services demand behavior of the citizens. As an example, the study made by Mogaji & Jain (2020) and Richter (2020) discusses how employees adjust their service delivery based on purely digital and virtual workplace and how educators and learners cop up with the requirements of online learning environment in order to fight against the spread of COVID-19. These arguments further strengthened by several other studies that include Caligiuri, De Cieri, Minbaeva, Verbeke, & Zimmermann, (2020) and Mogaji & Jain (2020). They emphasized that it is unsurprising to expect e-learning will become a more common avenue for education in the post-COVID period, as well remote working. The collaboration may focus on issues regarding how to scale-up the best practice of South Korea's ICTs adoption for different online services to adapt for African countries' such as how successfully migrate to online working environment that may include on how to develop effective leadership, clear guidelines, and real commitment to secure organizational networks and data, maintain employees' productivity and creativity in situations where employees are not used to work online before the pandemic environment, lack appropriate devices and tools.

4. Developing Strategy on how to Develop Consumers and Employees' Digital Skill

In attempting to contain the virus, countries have established mobility restrictions and, in some cases, lockdowns, which have fundamentally disrupted the functioning of society and the economy. Due to this, the status of broadband especially e-government and online services are switching from an amenity to a necessity, as they become not only the main way to access information and services, but also one of the only remaining alternatives for economic, educational, and leisure activities as well as for social interactions to take place. The transformation of global business and public administration toward online services' delivery system is already accelerating because of the COVID-19 pandemic, as businesses and even NGOs struggle to survive. Those that survive the shakeout will be those that are expected to have best preparedness for adopting digitalization by developing their manpower digital skill.

However, not all consumers are equals in terms of access to networks or connected devices, or when it comes to the skills required to navigate digital services optimally. As investigated by Beaunoyer, Dupere, and Guitton (2020) digital inequalities especially digital skill divide represents a major risk factor of vulnerability for unemployment, exposure to the virus itself, and for the non-sanitary consequences of the crisis.

Therefore, the development of digital skills is an important part of building resilience to economic and social shocks like those presented by the COVID-19 outbreak. For effective consumers and employees' successful digital skill development the social support network factor for example female social media penetration has significant contribution in multiple ways. Van Lancker et al. (2020) study evidence that having assistance in the form of recommendations or advice from more experienced Internet users when problems arise would increase readiness for learning digital skill. It implies that the exposure to technologies in the social network raises the likelihood to learn and adopt online services. Hardill & O'Sullivan (2018) and Cruz-Jesus et al. (2016) further endorsed that digital skill is often needed to access online services, support and information provided by governments, businesses, healthcare service centers or schools and universities higher education institutions. Such initiative will have ample of opportunities to be successful if they consider learning from best performing counties in the industry as an example South Korea.

5. Sustainable Digital Transformation: Post COVID-19 to Tackle the Sustainable Development Goals (SDGs) of African Countries

Moving forward, the ICT mainstreaming scenario has three potential opportunities to make a definite contribution to the tackling of SDGs of Africa. First, through mainstreaming ICT, we have shown how the collaboration initiatives could take the lead in exploring ways of realizing the full potential of digital technologies in contributing to building a sustainable society (Ismagilova, Hughes, Dwivedi, & Raman, 2019). Second, as we continue to fight the pandemic while preparing for SDGs, it is equally crucial that we mitigate the unintended negative privacy violation and social consequences brought about by the adoption of digital technologies. Third, as the pandemic continues to evolve, COVID-19 is changing the way we live and work and we therefore urge the African leadership to play a more active role in addressing the SDGs agenda set forth by the UN. If we explore the potential value of the concept and practice of digital sustainability, ICT will have practical effects towards achieving SDGs (Pan & Pee, 2020).

The report of ITU (2020) verify that even though Africa has big potential of ICT that include mobile network coverage of 89.4 percent and 3G mobile broadband network coverage of 79 percent, the adoption rate is far below the capacity, 3G mobile broadband subscription is 34 percent and the internet usage is only 28.2 percent. This demand gap needs to be promoted by bridging the digital skill gap and awareness creation because the divide has serious constrains on the use of telemedicine, mobile health initiatives – or m-health, remote working and other digital responses required to fight COVID-19. African countries need more than ever to strengthen their ICT sector development and use their potential through putting in place conducive ICT enabling environment that may include comprehensive policy framework and community skill development.

6. The Association of COVID-19 Pandemic with Population Density and Level of Urbanization

The pandemic has been primarily focused on industrialized, high- and middle-income countries, and the eventual march towards low-income countries of Africa and Asia seems inevitable. No aspect of human society is untouched by this pandemic. The unprecedented strain on health systems along with job loss, business closures and the economic downturn has become a worldwide phenomenon. Hence, there is an urgent need to explore the differences in epidemiological measures and demographic distribution of COVID-19 between different countries. This would include a difference in progression of cumulative and fatality cases whether it depends on the population size, density and urban population (Jordan et al., 2020; Yang et al., 2020).

To the best of our understanding, the virus is primarily spread through physical contacts of people including potentially, in indoor crowded poorly ventilated settings. Therefore, we are developing hypothesis urban areas are more vulnerable to rural areas because of the density of the settlement.

7. Contribution of this Study

As the COVID-19 pandemic marches exponentially, evidence based empirical research is of high importance to analyze the current situation and guide intervention strategies. This study analyses the empirical data of COVID-19 cumulative cases in relative to urban population of the 46 sample from 54 African countries, representing 85.2 per cent of the total. The dataset allowed us to perform pooled and interaction effect model analyses, revealing ICT policy variables that have significant effects in managing coronavirus. This study is unique because such empirical based analyses have not yet been studied in the existing research that we tried to review, mainly because such a dataset as we constructed is just under formation, and we believe that the research outcome of this paper would help policy makers of African countries to devise various policy mixes suitable for their respective countries.

Thus, the main objective of this study is to describe the situation of African countries and aim to propose ICT based policy for the future course of action that will help fighting against the pandemic. We have also tried to identify the best performing countries (South Korea) that tackled the COVID-19 with the help of ICT to be taken as benchmark.

8. Why do we Propose Korean Practices as the Benchmark for African Ehealth Services Delivery System Reset?

Korea was able to successfully flatten the curve on COVID-19 in only 20 days without complete lockdown, which has had a huge impact on the country's economy, but maintaining a balance between quarantine and social distancing on one hand and continue with economic activities and production on the other was the secrete of solving the challenge. To this end, ICTs played a vital strategy addressing the challenge. Mobile devices were used to support early testing and contact tracing. Advanced ICTs were particularly useful in spreading key emergency information on novel virus and help to maintain extensive 'social distancing'. The testing results and latest information on COVID-19 was made available via national and local government websites. The government provided free smartphone apps flagged infection hotspots with text alerts on testing and local cases.

The Government of the republic of Korea prepared about its report on the title: "Flattening the curve on COVID-19 - How Korea responded to a pandemic using ICT" and made it available for public and declared that the government is ready to share its best practices for countries specially developing one. In the report it is noted that countries expected to make best efforts to turn the crisis into opportunity and make the best use of the cutting- edge ICT technologies in forefront of fighting against COVID-19. We should also work together by making every information gathered on COVID-19 readily available to all. Any information on fight against COVID-19 is a public good, and it should be provided quickly to everyone in need and in their fight against COVID-19, as they must act and response most quick and swift manner. In this pandemic we are all connected, any wisdom and experience must be also shared quickly and fairly.

Hence, in the study, we do not present the details of the practices of how Korean government used the latest ICT against COVID-19 since it is available for public and can be referred.

III. Research Hypothesis, Data and Models

1. Research Hypotheses

The hypotheses set in this paper are testing the effects of explanatory variables used in the existing studies on COVID-19 cumulative confirmed cases in relative to the urban population across African countries. These set of hypotheses are summarized in 〈Table 1〉, is to reconfirm whether the signs of determinants of curtaining COVID-19 spreads are congruent with those of the existing conceptual studies. In order to attain the goal, this paper run regression and interacts the variables online service index with the telecommunications infrastructure development index.

2. Data and Descriptive Statistics

We took initially the 54 African Union member states as population of the study and

Table 1. Explanatory variables of existing COVID-19 cases ICT-based prevention studies

Explanatory variables [Dependent variable: COVID-19 cases relative to urban pop.]	Expected signs
Urbanization rate from total population	Positive
E-Government services index	Negative
Online service index	Negative
Telecom infrastructure index	Negative
Universal health coverage index	Negative
Mobile 4th generation network coverage	Negative
Electricity coverage rate	Negative
Female social media penetration divide	Positive
Female schooling gap	Positive
Applications on local language rate	Negative
Server security index	Negative
Tertiary Education enrollment rate	Negative

dropped a few of them if cumulative confirmed cases, the dependent variable[2], are less than 1000 or even though more than 1000, if their data are incomplete in 2020 year of study period for the independent variables, 46 member states data are available but 8 countries among them are excluded because data for the independent variables are not available for them. We therefore collected a full cross-sectional dataset of 46 African countries and the data of South Korea for analysis. 〈Table 2〉 shows variables, their definitions and measurements, and abbreviations used in the analyses.

As stated early, this paper focuses on finding out significant policy variables on prevention and control of COVID-19 infection cases and also considers that policy effects are reflected in the joint policy variables interactions, not just in the unique

2) The dependent variable is the total of confirmed cases to the total of the countries' urban population because cities are densely populated and vulnerable to COVID-19 spread through people's contact.

policy variable intervention. The COVID-19 confirmed cases of the countries in relative to their urban population is shown in [Figure 1], and Appendix 1 lists of all sample countries in alphabetical order with their seven key variable data:

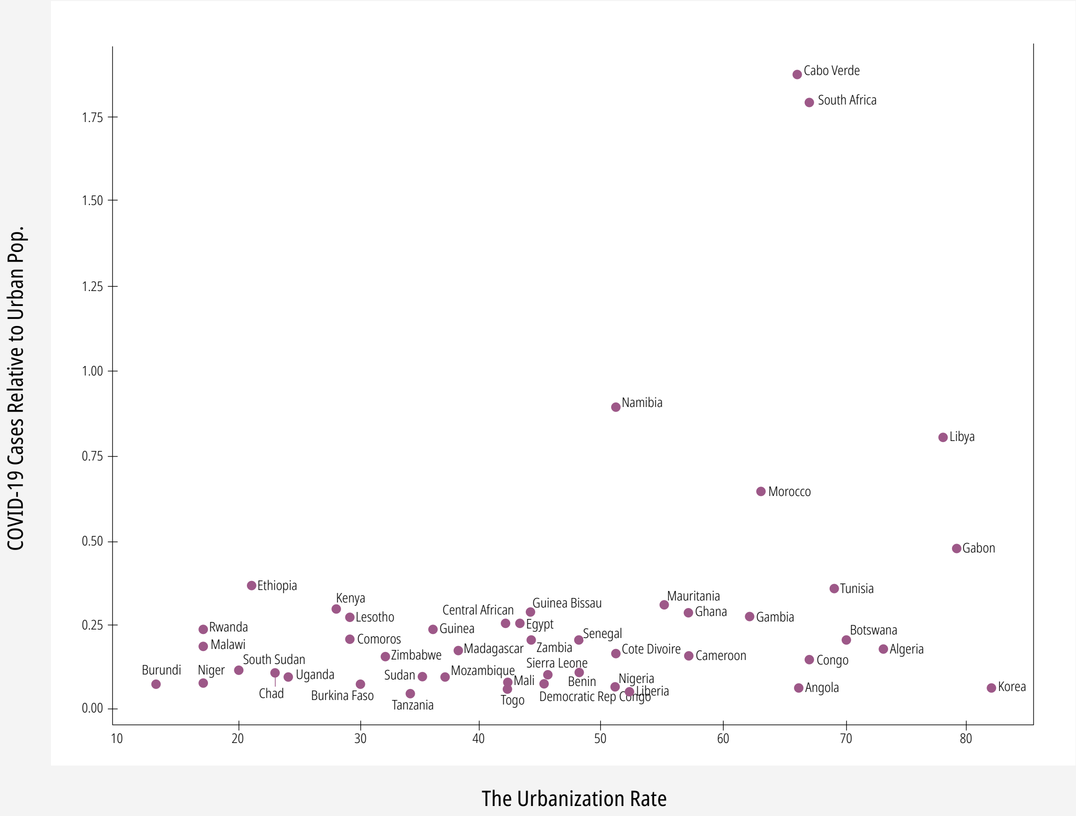
Descriptive statistics of 13 explanatory variables are summarized in Appendix 2. From [Figure 1], and Appendix 2, we can notice a few characteristics of the sample observations of the study. Due to the behaviour of the COVID-19 densely populated

Table 2. Definitions of Variables, Measurements and Abbreviations

	Variables	Definition	Abbreviation
1	COVID-19 cases	COVID-19 cases to total urban population percentage.	UPerCovid
2	Urbanization	Urbanization rate which is the urban people to the total population	PerUrban
3	E-Government dev’t index	A weighted average of normalized score of online service index, telecom infra index and human capital index	E-Gov
4	Online service index	The level of national online services index that include whole-of-government, e-participation, multi-channel service delivery, mobile services, usage uptake, and digital divide.	OnlineServ
5	Telecom infrastructure index	Estimated internet, fixed and mobile telephone, fixed and mobile broadband users per percent in the last 3 months	TelecomInfra
6	Universal health coverage index	The index is computed using geometric means of 14 tracer indicators to the scale of 0 to 100	UHCI
7	4G	Mobile 4th generation network coverage to the total population in percent	M4G
8	Electricity coverage rate	Electricity coverage percent to the total population	Elec
9	Female social media penetration divide	Female social media penetration percent gap in relative to male	GendgSMP
10	Female schooling gap	1Female school enrollment percent gap in relative to male	Gendschooling
11	Applications on local language availability rate	1Percent of the population who gets applications on their first language	Ap2Lang
12	Server security index	Number of secured servers per 1 million people	Cyberse
13	Tertiary Education enrollment rate	Tertiary education enrollment percent from the total population	Etertiary

Data sources: 1) from https://covid19.who.int/; 2), 6) and 8) from https://datacatalog.worldbank.org/dataset/UHC; 3), 4) , and 5) from https://publicadministration.un.org/ ; 7), 9), 10), 11), 12), and 13) from GSMA Mobile Connectivity Index https://www.mobileconnectivityindex

Figure 1. The Effects of Urbanization on COVID-19 Confirmed Cases



settlement (high Urbanization) shows direct correlation with total confirmed cases of coronavirus except few countries as an example South Korea which have low total confirmed cases in relative to high urbanization. This implies that Korea has developed successful prevention and control measurements and that is one of the reasons to be selected as benchmarked country. High universal health coverage has not shown the expected negative correlation with the COVID-19 confirmed cases instead its role might be positive for confirmed cases recovery that may need further investigation.

3. Research Models

To capture the statistically significant determinants of the prevention and control of COVID-19 cases across African countries, we first applied a pooled regression model, shown in Eq. (1) to the dataset that we introduced in the subsection above. The pooled model assumes that all African countries included in the analysis are random samples from the same population and ignores differences among countries. In other words, the pooled model implies that all variables affect the same effects on prevention and control of coronavirus across the countries. Further, prevention and control of COVID-19 cases with the help of ICT need to have combination of determinants factors instead of the individual impact factors and to capture such multifaceted determinants we used models with interaction terms. Hence, in order to capture the interaction terms, this paper added interaction term, β_{13}, interaction model, as shown in Eq. (2). Online service availability for example could be an important factor affecting the prevention and control of coronavirus if there is broadband network in the area. From the coefficients of interaction terms, β_{13}, we can separate out the differential effects of online services availability on COVID-19 cases prevention.

Pooled model: $y_j = \beta_0 + \sum_{i=1}^{12} \beta_i \chi_{ij} + u_j$ (1)

Interaction model: $y_j = \beta_{0j} + \sum_{i=1}^{12} \beta_i \chi_{ij} + v_j$ (2)

where y is COVID-19 total confirmed cases relative to urban population of the countries, β_0's intercepts, χ_{it}'s explanatory variables, β_i's the coefficient of the interaction terms, u_{jt} and v_{jt} disturbances, independently and identically distributed with mean zero and variance σ^2. We used all 12 explanatory variables and a constant in the pooled model, and additionally interaction terms.

IV. Results of Analyses: Effects on COVID-19 Confirmed Cases Relative urban Population: Pooled and Interaction Term Effect Models

We ran a pooled and interaction term effect models with a statistical package, R (version 3.4.3), to check if the coefficients of regression models are statistically significant and are congruent with expected signs in 〈Table 1〉. The regression results of the pooled and interaction terms effect models are presented in 〈Table 3〉. Before running the

Table 3. Regression Results of Pooled and Interactions Effect Models

Explanatory variables	Pooled model			Interactions effect model		
	Coefficients	t-value (p-value)		Coefficients	t-value (p-value)	
Intercept	-0.714	-3.01	(0.005)***	-0.476	-2.19	(0.036)**
PerUrban	0.008	2.06	(0.047) **	0.004	1.10	(0.279)
E-Gov	-2.719	-3.27	(0.002)**	-3.083	-3.68	(0.001)***
OnlineServ	1.912	4.18	(0.000)***	2.730	5.36	(0.000)***
TelecomInfra	0.965	1.52	(0.137)	2.110	2.34	(0.025)**
UHCI	0.019	3.01	(0.005)***	0.011	1.82	(0.078)*
M4G	-0.003	-1.78	(0.084)*	-0.003	-1.71	(0.097)*
Elec	0.002	0.67	(0.505)	0.002	0.62	(0.541)
GendgSMP	0.002	0.87	(0.388)	0.002	1.01	(0.319)
Gendschooling	0.006	2.22	(0.034)**	0.006	2.14	(0.040)**
Ap2Lang	-0.002	-0.93	(0.357)	-0.003	-1.37	(0.181)
Cyberse	-0.007	-2.88	(0.007)***	-0.009	-3.44	(0.002)***
Etertiary	-0.015	-2.97	(0.005)***			
OnlineServ*TelecomInfra				-2.360	-2.73	(0.010)**
Model significance	Degrees of freedom: 34 F-statistics: 6.56574 (p-value: 0.0000) Adjusted R2: 0.59216			Degrees of freedom: 34 F-statistics: 6.25831 (p-value: 0.0000) Adjusted R2: 0.57837		

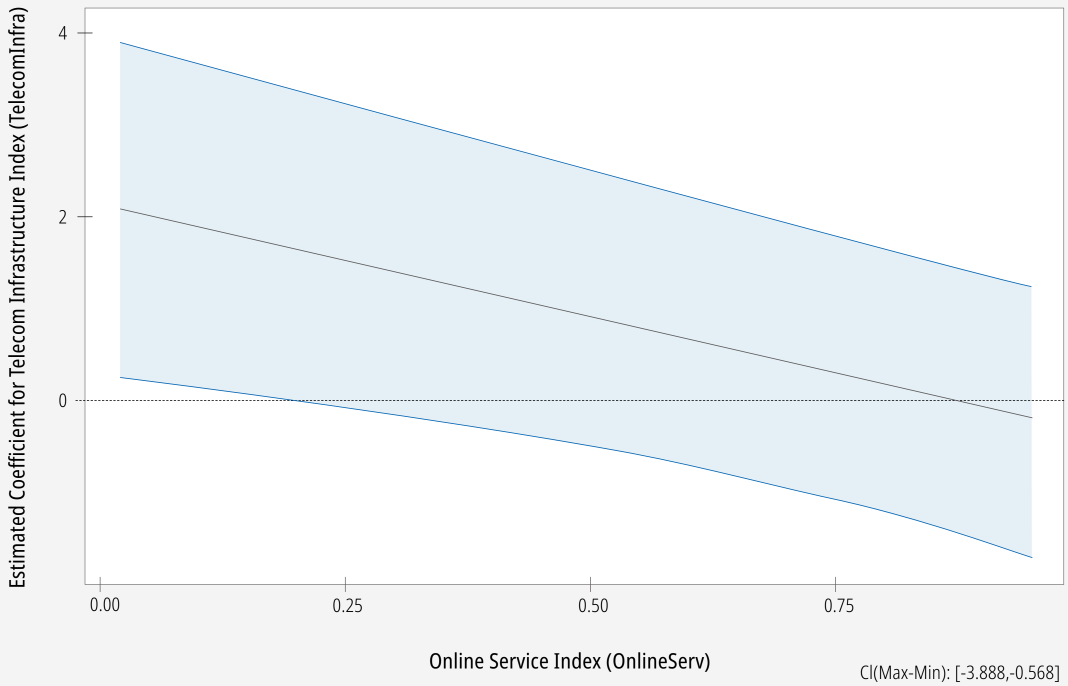
Note: *0.1, **0.05, and *** 0.01 are statistically significant level of rejecting the null hypotheses and accepting the alternative hypotheses developed on <Table 1>.

regressions, we calculated pairwise correlation coefficients among explanatory variables and presented the correlation coefficients in Appendix 3, which reveals that multicollinearity is not likely to be serious problem because the max of pairwise correlation coefficients is 0.8276 and only one value are in between 0.8 and the max. Surely, pairwise correlation coefficients are neither necessary nor sufficient for avoiding multicollinearity problem. Therefore, we calculated variance inflating factor (VIF) for the pooled model and found that only one variable, e-government, has a VIF of 10.86 and all other variables have VIF values far less than 10.[3] This implies that multicollinearity is not a serious problem in these regression analyses (Gujarati & Porter, 2009, p.340).

The results of the pooled model show that the coefficients of 8 variables of 12 excluding constant are statistically significant five at 1%, two at 5%, and one at 10% level of significance. Among those statistically significant explanatory variables five variables that include e-government service index (EGov), mobile 4G broadband internet (M4G), bridging female schooling divide (Gendschooling), secured server (Cyberse), and Tertiary education (Etertiary) turned out to be negatively affecting the COVID-19 confirmed cases (helping to prevent and control the pandemic). Urbanization rate (PerUrban), Online service index (OnlineServ), and Universal health coverage index (UHCI) were found to have positive relationship with COVID-19 confirmed cases, implying that the results are contradicting our hypotheses. However, there is interaction terms between the network infrastructure index and online services index, there is negative regression result which means instead of their individual presence, the variables would have synergetic effect when they are available concurrently as depicted the result in the [Figure 2] below.

3) VIF values: PerUrban (3.67), EGov (10.86), OnlineServ (6.56), TelecomInfra (6.76), Elec (4.10), UHCI (5.56), M4G (1.78), Gendschooling (2.89), GendgSMP (2.31), Ap2Lang (2.25), Cyberse (2.51), and Etertiary (4.79).

Figure 2. Estimated Coefficient for Telecom Infrastructure Index (TelecomInfra) on Urban COVID-19 Cases % by Online Service Index (OnlineServ)



V. Discussions and Implications of the Study

This study will shed light on the challenges of African telecommunication infrastructure to effectively deliver digital health services and possible ways for improvement. In Africa, the technology, media and telecommunications sector were expected to attract high value investments in 2020 with many telecommunications companies seeking to expand infrastructure, including the opportunities opening-up for the booming ehealth and other online services. Such initiatives may include encouraging the use of the digital payments over cash payments. At the same time, there are also opportunities, as the ICT industry has been providing solutions to contain the spread of COVID-19 through various applications and services, including the increasing use of communications technology for other basic service delivery and online learning.

To this aim, it underpins the importance of developing digital policy parameters related to ehealth especially tackling COVID-19 pertinent to ehealth service platform development, digital skill improvement, female social media penetration and schooling affirmative actions, e-government and online services improvement, improving secured mobile broadband internet and managing cities appropriately. Moreover, the paper promotes the mapping of the information and communications technologies (ICTs) infrastructure for emergency situations to help identify how policies and regulation made it difficult to harness ICTs to respond to the COVID-19 emergency across the continent and thus ensuring flexibility and continuous reform of the regulatory and policy framework in the future by collaborating with best-performing countries like South Korea.

For Africa effectively to deliver digital health services and other pertinent digital services, its telecom infrastructure services must be improved as prerequisite. African countries need more than ever to strengthen their ICT sector development through putting in place enabling legal and regulatory frameworks relating to cybersecurity, personal data protection and privacy, digital payments and supporting the growth of financial technology startups.

Despite in our regression model the identity registration has not shown statistically significant result, according to World Bank figures, more than 40 per cent of Africans lack an official identity papers on the part of many people in African countries, in particular digital identities, hampers the ability of Governments to identify and distribute appropriate welfare benefits and services for those affected by the disaster. Hence, digital ID systems are a critical mechanism that provide quantifiable benefits for such initiatives and African governments expected to ensure having a digital ID while mitigating the risks.

In the above discussion, we propose how initiating collaboration between African

Countries and South Korea can contribute to tackling the COVID-19 pandemic, and then enabling the achievement of the SDGs. As shown in the pandemic, the complexity, uncertainty, and difficulty for evaluation might lead to various multifaceted challenges in different fields (Gümüsay & Haack, 2020). This section turns to implications for this partnership study. We advocate two strategies that could help to yield contributions to the challenges.

First, we advocate that the issues emerged from tackling COVID-19 pandemic can be addressed by collaboration efforts. The issues highlighted are closely related to other SDGs. Therefore, an in-depth knowledge to respond to solving SDGs and to mitigate the unintended consequences of dealing with the issues would be crucial. For example, the COVID-19 pandemic has brought challenges to areas such as healthcare, governance, education, employment, and socio-economic activities. In this study, we identified what significant policy determinants should be given more focus that may include ehealth platform, digital skill development, female schooling and social media penetration inclusion, e-government services improvement, joint online service and network intervention, and secured broadband coverage when African governments are addressing the challenges.

Second, we suggest applying the principles of win-win scenario while implementing the initiatives. In addressing the identified challenges with the strategies, we recommend detail feasibility study is paramount to take the context of the respective countries and have appropriate adaption strategies. For example, when addressing the issues of e-government services delivery system development, responsible institutions of ICT are encouraged to adopt the ‘Stakeholder Involvement’ principle to engage various institutional entities in the development and implementation process.

VI. Conclusion

African countries that have made significant progress with their telecom sector investment, which in turn led them to the wide spread availability of broadband infrastructure, that should be able to reap the benefit in fighting COVID-19 and then helping them to recover their economy counter to the shock. According to the European Investment Bank study (2020) there are some innovative digital initiatives identified as offering easy to implement and affordable solutions include a telephone based application for contact-tracing in Kenya, drones spreading messages in rural areas in Ivory Coast and delivering samples to medical laboratories in Ghana and Rwanda, a self-diagnosis application available in 15 African countries. Moreover, there are an online learning platform in Tanzania for students and teachers in secondary schools, a platform to organize and monitor food and non-food distribution to the poorest people, as well as a system to map the effects and responses to the COVID-19 pandemic through social media data.

Having these positive initiatives and ICT existing potential, a lot could also be done further to ensure African citizens and students to enable benefitting from online healthcare service delivery and learning as is happening in developed countries. Even though there are several initiatives in many African countries in these respects, we have not read integrated and sustainable system development which can be taken as benchmark within the continent to be scaled up. To that end, South Korea has published the first version of Korean experiences on preventing and controlling COVID-19 cases highlighting the Korea's national responses to the pandemic and the author proposes this could be taken as initial concept for the collaboration between South Korean and African member States through the African Union (AU) as umbrella. It is trusted that this will help in responding to COVID-19 and having resilience for the economic shock using digital technologies. Therefore, AU can initiate its ICT policy development strategies which has been identified in this study to support member States in the formulation and implementation of digital technologies with a view to accelerating the socioeconomic transformation of Africa.

Whereas, as people of Africa and less developed countries suffer from the devastating impacts of the COVID-19 pandemic on their lives and economies, international communities are expected to make contributions to the fight against the current pandemics and the attainment of SDGs ever more than any time before. In order to act with this direction, contextualization of the industry for these emergency situations will be required by adapting the policies to respond to the COVID-19 emergency cases across the continent. To initiate such collaborations projects, there should be evidence based plausible plans appealable to the supporting countries. We hope this research provides insights about the importance of the collaboration with identified policy recommendations.

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Appendices

Appendix 1. The List of All 46 Sample Countries and Benchmarked South Korean in Alphabetical Order with Their Key Variables Dataset

Country	UPer-Covid	PerUrban	EGov	Online Serv	Telecom Infra	UHCI	M4G	Gendg-SMP	Cyberse	Etertiary
Algeria	0.17	73.00	0.30	0.07	0.19	78.00	67.20	38.18	26.20	51.37
Angola	0.06	66.00	0.33	0.35	0.14	40.00	50.00	49.55	9.70	9.34
Benin	0.10	48.00	0.20	0.14	0.15	40.00	45.00	18.56	48.50	12.27
Botswana	0.20	70.00	0.45	0.28	0.42	61.00	80.17	93.16	44.00	24.86
Burkina Faso	0.07	30.00	0.16	0.19	0.12	40.00	8.64	9.47	40.00	6.50
Burundi	0.07	13.00	0.23	0.15	0.03	42.00	25.00	20.20	8.70	6.05
Cabo Verde	1.87	66.00	0.37	0.46	0.36	69.00	5.00	85.93	5.10	23.62
Cameroon	0.15	57.00	0.28	0.22	0.13	46.00	53.40	59.06	43.20	12.76
Central African	0.25	42.00	0.08	0.01	0.04	33.00	8.64	18.35	3.60	7.45
Chad	0.10	23.00	0.13	0.14	0.05	28.00	12.00	8.16	9.80	3.25
Comoros	0.20	29.00	0.22	0.05	0.11	52.00	82.00	44.74	1.50	8.99
Congo	0.14	67.00	0.25	0.04	0.17	39.00	67.20	43.76	16.70	12.67
Côte d'Ivoire	0.16	51.00	0.22	0.19	0.17	47.00	55.18	34.80	45.60	9.34
DRep of Congo	0.07	45.00	0.19	0.09	0.08	41.00	34.00	31.83	0.80	6.60
Egypt	0.25	43.00	0.36	0.47	0.30	68.00	69.85	37.32	74.20	35.16
Ethiopia	0.36	21.00	0.27	0.53	0.05	39.00	69.94	19.41	27.80	8.11
Gabon	0.47	79.00	0.36	0.15	0.31	49.00	35.00	69.54	31.80	31.31
Gambia	0.27	62.00	0.24	0.20	0.20	44.00	30.96	17.48	28.00	14.26
Ghana	0.28	57.00	0.42	0.45	0.26	47.00	68.00	33.06	43.70	15.69
Guinea	0.23	36.00	0.12	0.09	0.09	37.00	8.64	51.79	19.10	11.56
Guinea-Bissau	0.28	44.00	0.18	0.11	0.08	40.00	62.00	16.98	5.50	12.42
Kenya	0.29	28.00	0.42	0.56	0.18	55.00	64.26	56.59	64.80	11.46
Korea	0.06	82.00	0.89	0.94	0.85	86.00	100.00	96.33	87.30	94.35

Appendices

Country	UPer-Covid	PerUrban	EGov	Online Serv	Telecom Infra	UHCI	M4G	Gendg-SMP	Cyberse	Etertiary
Lesotho	0.27	29.00	0.28	0.14	0.18	48.00	75.00	94.71	5.10	10.20
Liberia	0.05	52.00	0.23	0.24	0.10	39.00	48.94	55.60	20.60	16.92
Libya	0.80	78.00	0.43	0.11	0.43	64.00	40.00	45.00	20.60	39.02
Madagascar	0.17	38.00	0.24	0.22	0.05	28.00	61.89	66.70	19.60	5.35
Malawi	0.18	17.00	0.24	0.22	0.05	46.00	77.53	20.73	27.50	14.15
Mali	0.08	43.00	0.18	0.09	0.21	38.00	43.64	0.48	8.50	4.52
Mauritania	0.30	55.00	0.17	0.07	0.15	41.00	3.00	24.25	10.70	5.00
Morocco	0.64	63.00	0.42	0.64	0.34	70.00	78.00	29.70	42.90	35.94
Mozambique	0.09	37.00	0.23	0.20	0.10	46.00	12.44	38.58	15.80	7.31
Namibia	0.89	51.00	0.37	0.28	0.27	62.00	39.00	85.55	12.70	22.89
Niger	0.07	17.00	0.06	0.07	0.06	37.00	7.20	0.10	9.40	4.41
Nigeria	0.06	51.00	0.33	0.41	0.20	42.00	45.20	45.43	65.00	21.31
Rwanda	0.23	17.00	0.34	0.46	0.11	57.00	79.00	19.59	69.70	6.73
Senegal	0.20	48.00	0.33	0.38	0.20	45.00	73.65	26.78	30.50	12.76
Sierra Leone	0.07	42.00	0.16	0.12	0.12	39.00	42.00	45.28	13.80	13.67
South Africa	1.79	67.00	0.45	0.56	0.38	69.00	85.70	76.17	45.20	22.37
South Sudan	0.11	20.00	0.18	0.12	0.05	31.00	2.00	0.10	6.50	3.71
Sudan	0.09	35.00	0.25	0.22	0.19	44.00	30.96	50.00	29.40	16.92
Tanzania	0.04	34.00	0.55	0.57	0.09	43.00	28.00	35.99	64.20	4.01
Togo	0.06	42.00	0.31	0.32	0.10	43.00	12.44	11.64	8.70	14.52
Tunisia	0.35	69.00	0.47	0.52	0.35	70.00	80.00	63.58	53.60	31.75
Uganda	0.09	24.00	0.36	0.50	0.11	45.00	63.88	48.17	62.10	4.84
Zambia	0.20	44.00	0.35	0.37	0.12	53.00	49.10	59.79	43.60	29.55
Zimbabwe	0.15	32.00	0.35	0.26	0.22	54.00	40.00	45.88	18.60	10.01

Appendices

Appendix 2. Summary Descriptive Statistics of Key Variables

Variable	Obs	Mean Std. Dev.	Min	Max
UPerCovid	47	.2782979 .3767499	0.04	1.87
PerUrban	47	45.46809 18.51187	13	82
EGov	47	.2968085 .1404822	0.06	0.89
OnlineServ	47	.2759574 .1986986	0.01	0.94
TelecomInfra	47	.1842553 .1454191	0.03	0.85
UHCI	47	48.40426 13.13419	28	86
M4G	47	47.24787 26.70482	2	100
Elec	47	49.57234 29.03158	4.33	100
GendgSMP	47	41.36277 25.5921	0.1	96.33
Gendschool~g	47	62.49298 23.85755	14.3	100
Ap2Lang	47	40.97574 24.89495	3	82.83
Cyberse	47	29.57234 22.3955	0.8	87.3
Etertiary	47	16.74974 15.84946	3.2513	94.34969
PopMil	47	27.90894 35.37833	0.55	195.88

Appendix 3. Pairwise Correlation Table for Explanatory Variables

	perur-ban	egov	on-line~v	tel-eco~a	uhci	m4g	elec	gendg-smp	gend-sc~g	ap2lang	cyberse	eterti~y
perurban	1											
egov	0.49	1.00										
onlineserv	0.19	0.83	1.00									
telecominfra	0.71	0.81	0.56	1.00								
uhci	0.54	0.75	0.57	0.79	1.00							
m4g	0.25	0.55	0.49	0.43	0.52	1.00						
elec	0.69	0.61	0.46	0.72	0.77	0.49	1.00					
gendgsmp	0.49	0.59	0.36	0.59	0.54	0.40	0.47	1.00				
gendschool~g	0.37	0.61	0.31	0.53	0.61	0.41	0.43	0.66	1.00			
ap2lang	0.57	0.55	0.49	0.56	0.60	0.30	0.66	0.41	0.31	1.00		
cyberse	0.17	0.66	0.74	0.46	0.47	0.50	0.41	0.24	0.30	0.42	1.00	
etertiary	0.65	0.73	0.49	0.85	0.80	0.43	0.66	0.49	0.48	0.55	0.45	1.00

Chapter 4

Methodological Issues to Analyze the Disseminating Patterns of COVID-19: Focusing on Gyeonggi Province in South Korea

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Abstract

COVID-19, which started in Wuhan, China, is spreading around the world, and spreading to local communities is progressing at a rapid pace throughout Korea. The central government and local governments jointly cope with infectious diseases based on laws enacted after the MERS in Korea. In particular, they are used for classification of confirmed cases according to the guidelines suggested by WHO. This study conducted a migration path analysis based on the migration path data of 237 COVID-19 confirmed patients in Gyeonggi-do from January to April 2020, and classified the confirmed patients by type according to WHO guidelines. By standardizing the places visited by the confirmed patient based on the KSIC, the purpose of presenting the characteristics of the route of movement of the confirmed patient and the direction of institutional supplementation in the future was suggested.

As a result of the analysis of the path of movement of the confirmed patients, home, community, medical environment, and other types are concentrated, and long-term protection facilities, prisons, and other types of transportation according to the WHO guidelines have not been found. the example types of'other' in the current WHO guidelines rather correspond to the situation in Korea, so it is necessary to develop the types of chapels, workplaces, and private gatherings as the main types and use them for future infectious disease response.

I. Introduction

COVID-19, which started in Wuhan in December 2019, China, spread to Korea since January 2020. The rapid spread of pandemic has brought various discussions in relation with this.

The clinical characteristics in relation with COVID-19 (Ahn et. al., 2020; Lee & Na, 2020; Yong, 2020) were first mentioned, the personal rule for blocking (Goh, 2020; Park, 2020; Lee, 2020a; Sun & Jin, 2020; Lee, 2020b), facility standard (Joe, 2020), personal health (Nam, 2020) and the treatment related social problems (Son, 2020) were discussed.

With this, the discussion in the various fields such as international trend (Park, 2020; Chen et. al., 2020) and future prediction (Yoon, 2020), the policy to overcome COVID-19 (Lee, 2020c; Lee et. al, 2020; Lee, 2020d; Jang, 2020; Han, 2020; Heo, 2020), statistics (Gang, 2020), local cases (Shin, 2020; Lee, 2020e; Ji, 2020), the impact on industry, economy (Shin & Yoo, 2020; Lee, 2020f; Ju et. al., 2020g) was made.

In case of Korea, it is necessary to remind COVID-19 was not the first infectious disease. There existed the times when big scaled infectious diseases such as SARS, MERS influenced to Korea, and through that period, the institutional supplementations have been made to cope with the infectious disease. As aforementioned, there have been various discussions in relation with COVID-19 but the most important thing is to understand the system has been set over the process to overcome infectious disease and to supplement the existing system to cope with the next infectious disease properly.

After experiencing MERS, Korea has enacted「INFECTIOUS DISEASE CONTROL AND PREVENTION ACT」and provided the institutional basis for the organic response by the government, local governments, and medical institutions upon the outbreak of infectious diseases. This law is evaluated as a very powerful system to the extent the existing “protection of personal information” laws doesn't apply in the event of national infectious disease.

This system functions effectively to the national response under the current situation of COVID-19. 「INFECTIOUS DISEASE CONTROL AND PREVENTION ACT」 allows the nation to identify the visit place, time, the means of transportation and the purpose of visit of the confirmed patients and to deliver the information to people when a person is confirmed as COVID-19. In addition, it allows the local test facility to test the sample of suspicious patients of each region. This allows to confirm and respond the infectious disease within the short time.

In addition, [COVID-19 response instruction (for local governments)] that specifies the information disclosure and type specific epidemiological investigation of local governments have enabled the government to mobilize administrative power to block nationwide infectious disease. In particular, [COVID-19 response guideline (for local governments)] suggests WHO guideline that can identify the person who contacted the confirmed patients, and the local governments respond the infectious disease based on this guideline.

Despite of this, since WHO guideline is based on the global common situation not Korean situation, it is necessary to improve it according to the characteristics of Korea. In addition, it is very important which case should be based in the process to improve the system related to infectious disease.

That's because it is not difficult to generalize and reflect the quarantine system in case of the region with poor accessibility or very small population even if the infectious disease spreads suddenly. Therefore, if supplementing the infectious disease related system in the future is the purpose, the representability and generality should be considered when selecting the target of analysis. In terms of case analysis region in Korea, Gyeonggi-do, a metropolitan area, is adjacent to China, the origin of COVID-19.

Gyeonggi-do is the hinder land of Seoul, a capital of Korea, the headquarters, branches, offices, research facility, industrial basis and residential dense regions are mixed. In addition, it is characterized by the region of which transportational networks connecting Seoul and each region such as road, ports and railways are very much developed than other regions. Considering the characteristics of metropolitan area where the majority of all the nationals are dense, the developed transportational network will make the blocking of disease spread harder. As the infectious disease is estimated to bring various problems such as social chaos and economic stagnation, this region should be examined definitely during the stage of establishing the quarantine policy for the infectious disease response.

Under these backgrounds, this study raises the following questions ① which characteristics does the routes of confirmed patients show? ② WHO guideline properly fits the situation in Korea?

The purpose of this study is to figure out the features by utilizing and analyzing the data of confirmed patients route by local government, and to suggest the application plan of WHO Guideline in Korea. This study investigates the status of confirmed patients in Gyeonggi-do in the early period of COVID-19 spread and suggests the research methodology for the purpose of this study. Based on the analysis result in IV, V concludes.

II. COVID-19 Outbreak Status

1. COVID-19 Outbreak Status in Gyeonggi-do

The analysis period of this study is from January to April 30, 2020, the numbers of COVID-19 confirmed patients in Gyeonggi-do are shown in [Figure 1]. From January when the confirmed patients occurred for the first time to February 23 when Central Disease Control Headquarters elevated the crisis stage to 'serious', Central Disease Control Headquarters collectively managed and announced the status of confirmed patients across the country. Accordingly, the confirmed patients in Gyeonggi-do were traced and announced from January 29.

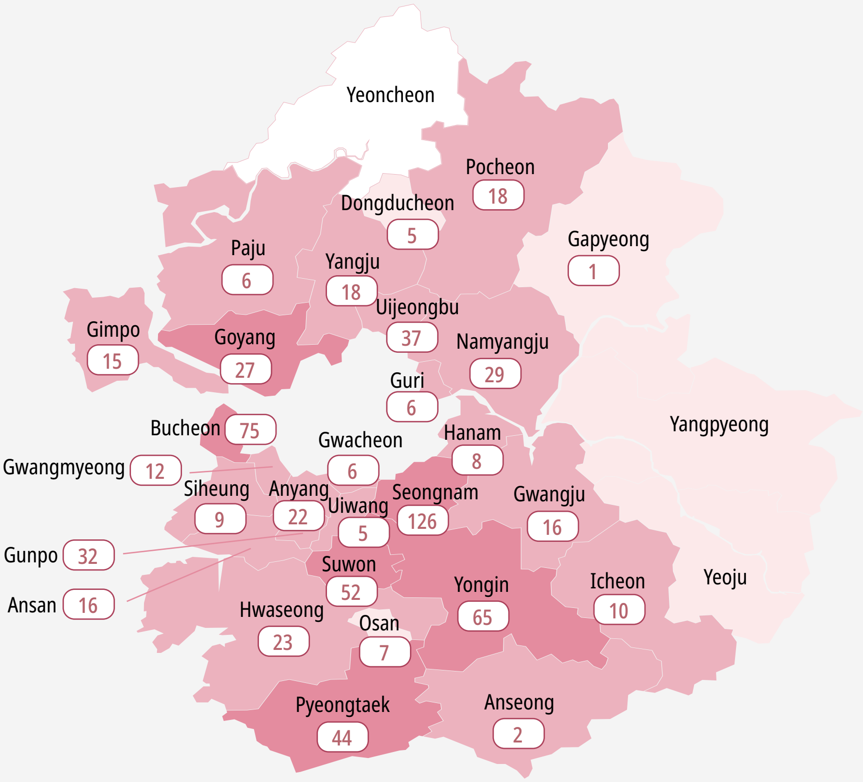
The first confirmed patients (No 1) in Gyeonggi-do is a person with Chinese nationality (35 years old) who was found during the quarantine at Incheon Airport on January 19, and No 2 confirmed patient was Seoul citizen who was found in the quarantine at Incheon Airport, was quarantined. The patients actually occurred in Gyeonggi-do were No 3, 4 confirmed patients who were later identified as Goyang and Pyeongtaek citizen, each of them are quarantined to Myeongji Hospital and Bundang SNU hospital. Afterward, No 2 confirmed patient was classified into the confirmed patient in Gyeonggi-do, later announced to be the resident in Bucheon with Chinese nationality. According to the announcement data on February 2, No 14 (Chinese nationality, woman, Bucheon) and No 15 (Suwon) were identified as the locals in Gyeonggi-do.

In Gyeonggi-do, the confirmed patients remained only 19 until February 22 before the elevation to 'serious' stage, the relatively stable management are conducted compared to Daegu, Gyeongbuk region where the numbers of confirmed patients skyrocketed. However, from February 22, there showed the sign to start regional spread, and the spread began in earnest due to contact with confirmed patients, staying in the same place, and family contact with confirmed patients.

Later, as the spread of the region progressed, 677 Confirmed patients in Gyeonggi-do had occurred by April 30. Seongnam had the largest number of 126 confirmed patients, followed by Bucheon with 75 people, Yongin with 65 people, Suwon with 52 people and Pyeongtaek with 44 people.

No outbreak in Yeoncheon-gun, Yangpyeong-gun, and Yeoju while Gapyeong-gun and Anseong-si had relatively few Confirmed patients with one and two. For 105 days since the outbreak of Confirmed Patients nationwide, the confirmed Patients in Gyenggi-do has exploded.

Figure 1. Status of COVID-19 confirmed cases in Gyeonggi-do (As of 0:00 April 30, 2020)



Source: Gyeonggi-do COVID 19 bulletin (gg.go.kr)

2. COVID-19 Response Policy

Identifying the route of confirmed patients is based on 「INFECTIOUS DISEASE CONTROL AND PREVENTION ACT」 which enacted since MERS. Article 34-2 (Disclosure of Information during Infectious Disease Emergency) clause 1[1] regulates to disclose the information during the issuance of a crisis warning in accordance with

1) Article 34-2 (Disclosure of Information during Infectious Disease Emergency) Ð Where the spread of an infectious disease harmful to citizens' health results in the issuance of a crisis alert of the caution level or higher prescribed in Article 38 (2) of the Framework Act on the Management of Disasters and Safety, the Minister of Health and Welfare shall promptly disclose information with which citizens are required to be acquainted for preventing the infectious disease, such as the movement paths, transportation means, medical treatment institutions, and contacts of patients of the infectious disease, by posting such information on the information and communications network, distributing a press release, etc.

Article 38, clause 2[2)]

In accordance with the issuance of a crisis alert by the Minister of Public Administration and Security, Director of the Korea Centers for Disease Control and Prevention is obliged to disclose information of the Confirmed patients, and each local government collects and routes of Confirmed based on 「[COVID-19 response guideline (for local governments)]」

WHO's 「Contact Tracing in the Context of COVID-19 Interim Guidance」 is applied to the routes of confirmed patients, inspection and information collection on the contact. However, in actual quarantine sites, WHO's guidelines are considered, but the scope of contact is determined according to the judgment of quarantine officers and epidemiological investigators. The routes of the confirmed patients are first identified through an interview, and prompt quarantine measures are implemented and the investigations such as GPS, DUR, and card usage history are conducted only when necessary according to the judgment of the municipal quarantine officer. The subject of disclosure is limited to patients with infectious disease, and information will be disclosed from 2 days before the occurrence of symptoms to the date of quarantine

In accordance with the place where the contact with the confirmed patients occurs (including the means of transport) and the epidemiological investigation results, the scope of disclosure of the routes of the confirmed patients will be determined by comprehensively considering the symptoms of the confirmed patients, whether the mask is worn, the duration of the stay, the exposure situation, and the timing of the patient. There are two reasons to identify and disclose information on routes of the

2) Article 37 (Issuance, etc. of Crisis Alerts) ① Where the head of a disaster management supervision agency identifies any sign of a disaster prescribed by Presidential Decree or expects the occurrence of a disaster, he/she may determine the risk level, possibility of occurrence, etc. and issue a crisis alert so that corresponding measures can be taken: Provided, That in circumstances falling under the proviso to Article 34-5 (1) 1, the Minister of the Interior and Safety may issue a crisis alert. 〈Amended by Act No. 14839, Jul. 26, 2017?
② Crisis alerts issued under paragraph (1) may be classified into attention, caution, alert, and serious level, comprehensively taking into account the seriousness of a disaster situation, such as the speed of development and possibility of expansion of disaster damage: Provided, That where the criteria for issuing crisis alerts of disasters are separately prescribed by any other statute, such criteria shall govern.

Confirmed patients. First, it is to identify information that the public needs to know in order to prevent such cases as routes, medical institution, and means of transport. Second, it is to control the situation, such as the outbreak situation, epidemiological investigation, disinfection of the visiting area and monitoring of the infected.

The recently revised [ENFORCEMENT DECREE OF THE INFECTIOUS DISEASE CONTROL AND PREVENTION ACT] provides methods and procedures for self-institutional treatment. The system will be improved to allow self-institutional treatment at the doctor's discretion, and if it is difficult to accommodate all infectious disease patients due to the outbreak of large number of infectious disease patients, etc., Treatment is allowed in a quarantine, nursing home or clinics installed and operated in accordance with Article 37 (1) 2 of the Act.

In the event of a Pandemic phenomenon and a large number of conformed patients, it is important to secure beds in the region, to identify available resources, and to ensure proper placement of medical resources. Since the resources are invested in the examination of suspected patients and treatment of confirmed patients, proper and effective quarantine management is required by identifying the routes of the confirmed patients. In particular, according to WHO's guidelines, that’s because 15 minutes of stay will be judged an important requirement, and special care will be more important for repeatable exposure areas.

III. Research Method

1. Data to be Analyzed

This study collected the information on the routes of each confirmed patient to analyze the routes of the confirmed patients. The data on the routes of confirmed patients for COVID-19 went through three change processes. First, from January to February

Table 1. Examples of WHO contact range

setting	Specific contact by setting
Household and community/ social contacts	· Face-to-face contact with a case within 1 metre and for >15 mins · Direct physical contact with a COVID-19 patient · Providing direct care for a COVID-19 patient in the home without proper PPE (Personal Protective Equipment) · Anyone living in the household
Closed settings, such as longterm living facilities, and other high-risk congregational/closed settings (prisons, shelters, hostels)	· Face-to-face contact with a case within 1 metre and for >15 mins · Direct physical contact with a COVID-19 patient · Providing direct care for a COVID-19 patient in the home without proper PPE · Sharing a room, meal, or other space with a confirmed patient · If contact events are difficult to assess, a wider definition may be used to ensure that all residents, especially high-risk residents, and staff are being monitored and screened
Healthcare settings	· Health care workers : any staff in direct contact with a COVID-19 patient, where strict adherence to PPE ha failed · Contacts exposed during hospitalization : any patient hospitalized in the same room or sharing the same bathroom as a COVID-19 patient, visitors to the patient, or other patient in the same room; other situations as dictated by risk assessment · Contact exposed duing ourpatient visits : Anyone in the wating room or equialent closed environment at the same time as a COVID-19 should be listed as a contact · Anyone within 1 metre of the COVID-19 patient in any part of the hospital for > 15 minutes
Public or shared transport	· Anyone within 1 metre of the COVID-19 patient for > 15 minutes · Direct physical contact with a COVID-19 patient · Anyone sitting within two rowsof a COVID-19 patient for > 15 minutes and any staff (e.g. train or airline crew) in direct contact with the case
Other well-defined setting and gatherings (place of worship, workplace, schools, private social events)	· Anyone within 1 metre of the COVID-19 patient for > 15 minutes · Direct physical contact with a COVID-19 patient · When events are difficult to assess, the local risk assessment may consider anyone staying in the same close and confirmed environment as a COVID-19 patient as a contact

Source : WHO (2020), Contact tracing in the context of COVID-19 Interim guidance

22, 2020, Central Disease Control Headquarters collected and announced the routes. Second, after the infectious disease crisis stage was upgraded to “serious” (February

23), each local government collected and announced information on confirmed patients and migration paths in the region. Third, a government guideline was issued to delete the data released from May 3 to May 2 on the ground of economic difficulties and personal information protection in the region. Accordingly, the collection of related data before May 2nd was restricted.

The collection and announcement of information on confirmed patients and routes by local governments was carried out with several characteristics. First, even if a person residing in A city tested positive at a health center in B city, the confirmed patient information is classified as B city's information, and the confirmed patient code is also assigned as B city's code. Second, the migration path only in jurisdiction was collected and announced.

When the confirmed patient at B city moves to city A, B city collects and announces only the movement route within the city B, and the routes at city A is collected and announced at city A. Third, there was a large deviation between the accuracy levels of the announced routes information of confirmed patients by local governments. In the case of C city, the name of the store where the confirmed patient moved and the travel time were disclosed in details relatively while the accuracy of D city information was relatively low, and the information between the residents of the community was more accurate.

This study considered the characteristics of announced COVID-19 confirmed patients information by each local government described earlier, and the information of Confirmed patient was collected at the information service bulletin board for each local government. The information collected is as shown in 〈Table 2〉.

In the process of collecting the confirmed patient's information, the confirmed patient who stayed at home and did not show a routes was excluded from the collection. In addition, even if Gyeonggi-do residents but if there is no routes in Gyeonggi-do or if they reside in other region, they were excluded from the information collection.

The information on the routes of each local government mainly consists of the date,

Table 2. Numbers of route

(Unit: ea, %)

Residence	Numbers of routes in	Proportion
Gangwon-do	15	0.6
Gyeonggi-do	2,563	98.2
Seoul	26	1.0
Incheon	7	0.3
Total	2,611	100.0

time and location of the visit, and the means of transportation, purpose of visit, and the relationship between contact is sporadically announced depending on the local government. This study mainly collected the important information and secured a total of 2,611 routes. The collected data showed residents of Seoul, Gangwon-do and Incheon who entered Gyeonggi-do and their routes was identified, and their 26, 15 and 7 migration paths were collected, respectively. In fact, the movement paths of residents in Gyeonggi-do were 2,583 which accounts for 98.2% of the total collected data.

2. Research Methodology

This study used Q-GIS (Quantum GIS), an open source GIS program to analyze the routes of confirmed patients. GIS is a tool that can analyze real world phenomena by integrating and managing spatial data and attribute data possessing geographic location information. A data table was created as shown in 〈Table 3〉 to visualize and analyze the visited location and routes as GIS data.

First, a total of 237 information on routes in the metropolitan area was collected from January 16, 2020 to April 30, 2020, and GIS point data was created through geocoding method that converts the address information of collected data into coordinate data. Geocoding is a method of converting address information consisting of text into

Table 3. **Table components and examples of confirmed patients data**

Metropolitan region	City, gun, gu	confirmed patient code	visit time	Name of visit place	Means of transportation	Address of visit place
Gyeonggi-do	Guri city	Guri○	2020-02-03 20:15	Gwangnaru Station	On foot	Gwangnaru Station, 237, Gwangjang-dong, Gwangjin-gu, Seoul
Gyeonggi-do	Gwangmyung city	Gwangmyung△	2020-03-03 11:40:00	□□internal medicine department	On foot	864, Haahn-dong, Gwangmyung, Gyonggi-do
Gyeonggi-do	Suwon city	Suwon□	2020-03-08 17:00:00	Incheon Airport	Aircraft (Return)	272, Gonghang-ro, Jung-gu, Incheon
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Source : created by author

geographic location information, latitude or coordinates. Transformed GIS point data (Spatial data) contains attribute data about the visit place and the visit time of each conformed patients.

Second, The routes of each confirmed patient was expressed with line after going through the work that converts point to line (points to line) in order to express the routes of each confirmed patient over time. The line was expressed with arrows according to the flow of movement, and the different color used for each confirmed patient so that the routes by confirmed patients could be visually distinguished.

Third, the routes analysis data of this study has been categorized in accordance with the contact range example criteria suggested by WHO (2020) contact tracing in the context of COVID-19 Interim guidance

It is necessary to figure out the characteristics of the confirmed patient's visit location information. This is because there are many cases that grant the different name even for places of the same type or industry. Therefore, this study conducted the standardization to identify the characteristics of the confirmed patient's visit place, and was reclassified and categorized the place visited by the confirmed patients according to the index of detail classification (5 digit)provided by KSIC (Korea Standard Industry Code) ver.10 (KSIC-10). For example, ◯◯hair salons visited by confirmed patients, △△hair shops were united and converted to 'Hair beauty shops' (96112), and ◯◯mart, △△supermarket are converted to 'Supermarket' (86102) by KSIC 10.

IV. Analysis Result

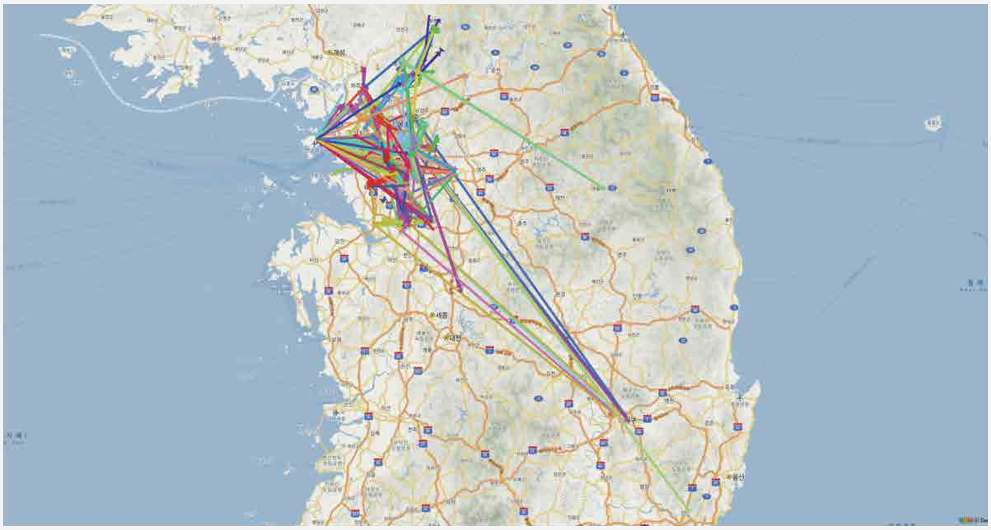
1. Routes Analysis

The routes of confirmed patients in Gyeonggi-do from January to April 2020 is as shown in [Figure 2]. It was analyzed that most of the confirmed patients moved within the metropolitan area, including Seoul and Gyeonggi-do. However, it was identified a few people in the earliest period have visited Daegu and Gyeongsangbuk-do.

In addition, as residents of Seoul, Incheon, and Gangwon-do visited Gyeonggi-do, routes within Gyeonggi-do were observed. When the area visited by the confirmed patient was standardized by KSIC-10, it was analyzed that the confirmed patient visited 109 site types. The place where the confirmed patients stayed the most was their home or family's house, and a total of 835 visits were recorded. In addition, the average routes per confirmed patients was shown to be 11. During the same period, in order to inhibit the spread of COVID-19, the government recommended refraining from movement and implemented social distancing campaigns, so many people stayed at home. Due to this reason, many people visited supermarkets (64 times), CVS (60 times), and coffee shops (40 times), which are relatively close from home.

In addition to Home, many places visited are medical facilities, there were 622 visits including general hospitals (247 times), public health operational business (179 times), and general clinics (86 times). pharmacies (77 times), general hospitals (27 times), and

Figure 2. Routes of confirmed patients in Gyonggi-do (whole)



oriental clinics (6 times). There are four reasons showing the high frequency of visiting medical facilities. ① to receive the treatment by misunderstanding cold-like symptoms as the symptom of COVID-19 before confirmation of COVID-19. ② to nurse families hospitalized in hospitals. ③ because the cases of mass infections of patients, doctors, and nurses appeared at Bundang Jesaeng Hospital and Hyosarang Nursing Hospital, which are medical facilities in Gyeonggi-do. ④ because the confirmed patients are transferred to the hospital after confirmation of COVID-19

The second most visited place for confirmed patients is the means of transportation.

Table 4. Visit place and frequency of confirmed patients

Categories of Business	Frequency
Home	835
General hospitals	247
Public health centers	179
Urban railway transport	98
General clinics	86

Categories of Business	Frequency
Retail sale of pharmaceutical and medical goods	77
Supermarkets	64
Convenience stores	60
Operation of airport	44
Coffee shops	40
General korean food restaurants	37
Bus stop	36
Christianity organizations	30
Specialized hospitals	27
Korean food restaurants specializing in meat dishes	25
Retail sale of bakery, flour confectionery and sugar products	21
Activities of call centers and telemarketing services	20
Pizza, hamburger and sandwich eating places and similar food services activities	19
Singing room operation	18
Discount department stores	17
Child day-care activities	16
Operation of car park or garage facilities	16
Other drinking places	15
Postal activities	14
General korean food restaurants	12
Domestic commercial banks	12
Oil stations for transportation equipment	12
Terrestrial broadcasting	11
Hair beauty shops	10
Other technical and trade schools	9
Combined facilities support activities	9
Korean food restaurants specializing in seafood dishes	9
Wholesale of paint	8
Other superstores with sales space of at least three thousand square meters	7
Application software publishing	7
Physical fitness facility operation	7
Dried seaweed rolls and other light food restaurants	6
Department stores	6
Western food restaurants	6
Oriental medical clinics	6

Categories of Business	Frequency
Retail sale of cosmetics, soaps and perfumery	6
Bus terminal	6
Manufacture of abrasive articles	5
Funeral homes and funeral related services	5
Train station	5
Motion picture theaters	5
Other retail sale in non-specialized stores	4
Manufacture of articles made of metal wires	4
Veterinary activities	4
Sale of new motor vehicles	4
Repair services of motor vehicles specializing in parts	4
Executive administration of reginal and local bodies	4
Sanatorium for the elderly with nursing care	3
Manufacture of industrial process control equipment	3
Pet funeral and care services	3
Inns	3
Research and experimental development on medical sciences and pharmacy	3
Japanese food restaurants	3
Chinese food restaurants	3
Computer game room operation	3
Resort condominium	3
Household laundry services	2
Golf practice range operation	2
Household laundry services	2
Take-out light food restaurants	2
Health insurance	2
Highway rest area	2
Manufacture of other new parts and accessories for motor vehicles n.e.c.	2
Heat treatment of metals	2
Other general accommodation and provision of accommodation with cooking facilities	2
Library and archives activities	2
Reading room operation	2
Courts	2
Real estate agents and brokers	2
Retail sale of books, newspapers and magazines	2

Categories of Business	Frequency
Interior design services	2
General repair services of motor vehicles	2
Retail sale of vegetables, fruit and root crops	2
Service activities incidental to rail transportation	2
Chicken restaurants	2
Other beauty shops	2
Wholesale of household furniture	1
Retail sale of watches and jewellery	1
Korean food restaurants specializing in noodle dishes	1
Retail sale of household electrical appliances	1
Activities of management consultancy	1
Other foreign food restaurants	1
Other general public administration activities	1
Singing room operation	1
Massage salons	1
Retail sale of stationery and artists' goods	1
Retail sale of fresh, frozen and other fishery products	1
On-line educational institutes	1
Turkish baths, sauna and steam baths, solariums	1
Wholesale of sports and athletic goods	1
Retail sale of meat	2
General subject educational institutes	1
Renting of motor vehicles	1
General repair services of motor vehicles	1
Exhibition, convention and trade fair organization agencies	1
Retail sale of side dishes	1
Other complex sports facility operation	1
Middle schools	1
Elementary schools	1
Dental clinics	1
Manufacture of toothpastes, soaps and other detergents	1
Retail sale of communication equipment	1
Retail sale of flowers and plants	1
Unknown	282
Total	2,611

There showed a total of 188 visits including the subway (98 times), the airport (44 times), the bus stops (36 times), the train station (5 times), and the bus terminal (5 times). However, in the case of the spread of COVID-19 through the means of transportation, there are only cases of flight attendants and passengers in contact with the aircraft, and there are no other cases within the analysis data of this study. The reason that there are no other confirmed cases in transportation other than the confirmed cases of aircraft may be due to the fact that the use of public transportation is not long, and that even if the confirmed patient uses public transportation for a long time, the contact time between the confirmed patient and others may not be long. In addition, since the majority of confirmed patients moved by their own car, it seems the frequency of using public transportation and contagion cases are less.

In addition, the cases of call centers located in Guro-gu, Seoul, and religious organizations such as Sincheonji and churches (River of Grace Community Church) are the representative examples of mass infection in Gyeonggi-do. This example can be classified into other types of WHO (such as chapel, workplace, school, private gatherings, etc.).

2. Analysis of Visit Place

The information on the visit place used when analyzing the entire routes of the Confirmed patient was standardized by the industry code of KSIC-10, and then matched with the WHO guidelines, it was summarized as examples of household and community, healthcare setting, and other (such as chapel, workplace, etc). In this section, the characteristics of each of the three types were analyzed.

1) Household and community

It has a characteristics showing high infection rate of COVID-19. It can be infected from other only by 15 minutes contact which is relatively short, if staying a long time within the same space without direct contact, the possibility of infection may be increased. This characteristic of COVID-19 acts as the most dangerous risk to household and local community.

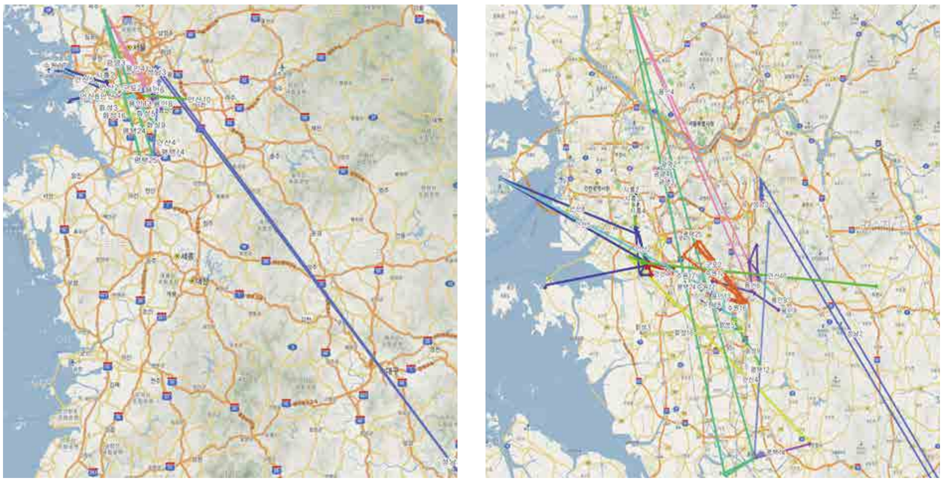
The reason why there are lots of inter-family infection is the increased opportunities in which family members reside together, and contact each other such as dialogue and eating. The cases of inter-family spread and the routes of confirmed patients are shown as [Figure 3].

It shows inter-family spread can occur nationwide. In particular, the period from January to April 2020 shows the trend of COVID-19 spread mainly in Daegu and Gyeongsangbuk-do.

During this period, a confirmed patient living in Daegu visited the house of family in Gyeonggi-do spreading COVID-19 to Gyeonggi-do. However, infections of COVID-19 are not clearly divided as inter-family spread. There were a number of complex cases, such as spreading to family members after infected in religious facilities and spreading to family members after infected by co-workers.

In terms of the natures of routes, the case of inter-family spread can be characterized by very long movement path. It is estimated many family members residing in other

Figure 3. The routes (nationwide) and the routes in Gyeonggi-do of inter-family infected confirmed patients



area moved far beyond the radius of life on the ground of 'winter vacation', 'visiting relatives'.

The cases confirmed to be inter-family spread were total 37 patients and total 384 routes were identified. The routes of average 10.4 was identified until the confirmed patients of this type are transferred to the hospital. It was confirmed they stayed in 45 business sites except for their house.

The place where confirmed patients stayed the most was at home, and 110 routes were identified, and they visited medical facilities such as general hospitals (30 times), public health services (24 times), general clinics (16 times), medicine and medical supplies retailing (14 times), general hospitals (11 times), etc.

In the inter-family spread, there were very limited cases of subways (13 times), airports (4 times), bus stops (2 times), and bus terminals (1 time), which are the types of transportation. It seems that's because they used their own vehicle as a means of transportation and, after arriving, only moved for a short distance on foot (Refer to 〈Table 5〉).

Table 5. Visit place and frequency of inter-family spread typed confirmed patients

Categories of Business	Frequency
Home	110
General hospitals	30
Public health centers	24
Supermarkets	20
Christianity organizations	17
General clinics	16
Retail sale of pharmaceutical and medical goods	14
Urban railway transport	13
Specialized hospitals	11
Singing room operation	9
Convenience stores	9
Coffee shops	9

Categories of Business	Frequency
Wholesale of paint	8
Child day-care activities	5
Operation of car park or garage facilities	5
General korean food restaurants	9
Operation of airport	4
Retail sale of bakery, flour confectionery and sugar products	4
Western food restaurants	4
Discount department stores	3
Combined facilities support activities	3
Household laundry services	2
Manufacture of other new parts and accessories for motor vehicles n.e.c.	2
Sanatorium for the elderly with nursing care	2
Discount department stores	2
Hair beauty shops	2
Bus stop	2
Veterinary activities	2
On-line educational institutes	2
Repair services of motor vehicles specializing in parts	2
Pizza, hamburger and sandwich eating places and similar food services activities	2
Korean food restaurants specializing in meat dishes	2
Korean food restaurants specializing in seafood dishes	2
Retail sale of watches and jewellery	1
Domestic commercial banks	1
Other retail sale in non-specialized stores	1
Dried seaweed rolls and other light food restaurants	1
Singing room operation	1
Department stores	1
Bus terminal	1
Oil stations for transportation equipment	1
General subject educational institutes	1
General repair services of motor vehicles	1
Chinese food restaurants	1
Executive administration of reginal and local bodies	1
Unknown	21
Total	384

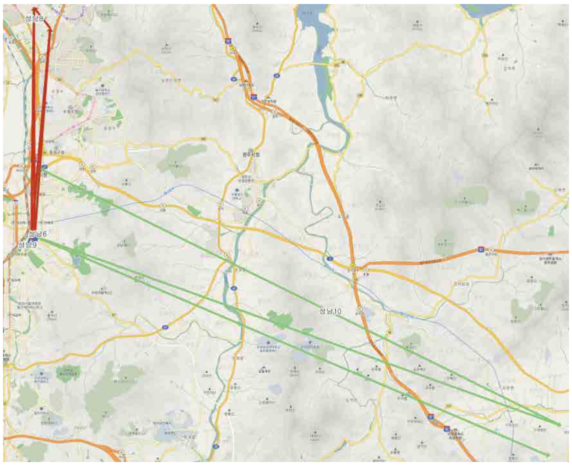
2) Healthcare setting

Bundangjaesang hospital in Seongnam city is the representative example of the healthcare setting type. It is the case of mass infection in which COVID-19 confirmed patients hospitalized in Bundangjaesang hospital and spreaded COVID-19 to the doctors, nurses and caregivers. In addition, it is the case of which mass infection range was expanded as the contact scope got wider due to movement of hospital ward

Healthcare setting type can be characterized by few routes outside the hospital because of many movement within the hospital building. The route can't be identified under the current system in which all of them conduct the self-quarantine before the expression of symptom and the routes are disclosed from 2 days before the expression of symptoms.

The confirmed patients of which route was identified in the healthcare setting case established as a data of this study are 3, showing the simple repetitive route of house->hospital->house. Visiting pharmacy in hospital and coffee house nearby was 1 time which is extremely low. In addition, in terms of transportation, Seohyun subway station is located nearby hospital, showing the subway is frequently used. However, it seems no infection case with the subway as a medium (Refer to [Figure 4]).

Figure 4. Routes of confirmed patient in medical environment type.



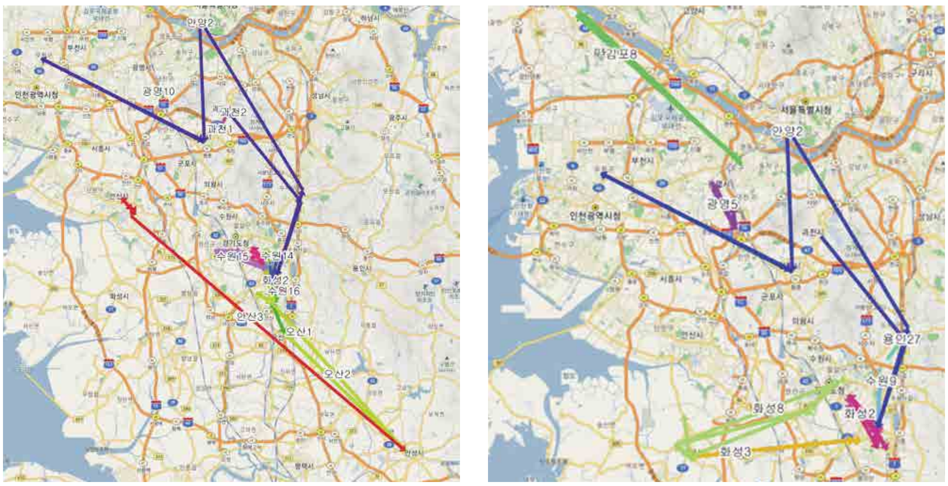
3) Other well-defined setting and gatherings (place of worship, workplace, schools, private social events)

The chapel and workplace classified as other type is an example showing the connecting link of spreading COVID-19 to Shincheonji, church and workplace. The left of [Figure 5] indicates the routes of confirmed patients related with Shincheonji and church. The feature here is the long route at the top of Figure, this is the path of Shincheonji confirmed patients. Compared to the church related confirmed patients move in the narrow space in the region, Shincheonji church members show the relatively longer route.

Next, the route of workplace related COVID-19 confirmed patients. The right of [Figure 5] is the workplace infection case but related with the left of [Figure 5] Shincheonji related confirmed patients. It shows Shincheonji related confirmed patients visited A company located in Hwaseong and spreaded COVID-19, and the patient infected in A company in Hwaseong moved to Hwaseong and Yongin.

In the event that the house is excluded, the chained convenience store (18 times) showed the highest frequency as a single business. In addition, business facility

Figure 5. **Route of Chapel (left) related and workplace related confirmed patients**



maintenance service business (9 times), christianity organization (8 times) which fall to other category (chapel, workplace) showed high frequency. The subway (8 times), bus stop (8 times) means of transportation, showed the similar level.

For other types, a specific person's infection is confirmed without symptoms, and the person classified as a contact is simultaneously tested for COVID-19, so hospital visits when the symptoms appear relatively low. Accordingly, it is estimated that the frequency of facilities and industries related to daily life appears high (Refer to 〈Table 6〉).

Table 6. Visit place and frequency of other (chapel, workplace) type confirmed patient

Categories of Business	Frequency
Home	57
Convenience stores	18
General hospitals	15
Public health centers	14
Combined facilities support activities	9
Supermarkets	9
Christianity organizations	8
Urban railway transport	8
Bus stop	8
Retail sale of bakery, flour confectionery and sugar products	3
Manufacture of industrial process control equipment	3
Retail sale of pharmaceutical and medical goods	3
Specialized hospitals	3
General clinics	3
General korean food restaurants	3
Other retail sale in non-specialized stores	2
Dried seaweed rolls and other light food restaurants	2
Reading room operation	2
Coffee shops	2
Household laundry services	1
Wholesale of household furniture	1
General korean food restaurants	1

Categories of Business	Frequency
Highway rest area	1
Retail sale of meat	1
Physical fitness facility operation	1
Pizza, hamburger and sandwich eating places and similar food services activities	1
Unknown	11
Total	190

The reason of COVID-19 in other type, first, even if people don't visit the chapel, the religious event in the region other than the chapel triggered the infection. Second, that's because the confirmed patient who participated in the religious event hided the fact of participation and visited A company.

3. Policy Implications

As a result of analysis, this study drew the following policy implications. First, because 5 types suggested by WHO Guideline are also observed in the visiting place and the routes of confirmed patients in Korea, the use of guideline is considered to be proper in a large context. There existed the types which fall to household and community, healthcare setting and other, there were no type for long-term living facilities, prisons, shelter types and transportation.

In particular, the spread through the transportation is found not to be discovered other than aircraft crew. In case of transportation, when considering many confirmed patients usually use their own car, even if the public transportation is used, the staying time with passenger is likely to be under 15 minutes, it is highly likely that the spread through transportation will not appear easily in the Korean situation. In addition, In the case of long-term living facilities, prisons, shelters, etc., the temporal characteristics of the early days of COVID-19 may have been reflected, but considering that these facilities are basically blocked from the outside, it is reasonable to exclude them from the future quarantine policy guidelines.

Second, it is necessary to classify and utilize 'other' type of WHO Guideline by reflecting the characteristics of Korean society. The 'other' type of WHO Guideline includes excessively many situations. It includes chapel, workplace, school and private gathering all, so it is necessary to consider each type acts as and important factor of mass infections.

'Shinchoenji' (chapel), the bar in Pyungtaek (private gathering) and Guro call center (workplace), which have been issued the most in the early period of spread, are the case of mass infection, and reported as a big news, and emerged as a serious social problem. It is necessary to arrange and respond with the guideline for each situation by setting them as main types rather than grouping and classifying as 'other' type simply because various types have the risk of mass infection.

Third, the effort to minimize the mass infection is needed. In case of Seongnam where the mass infections occur, the mass infection of medical personnel in Bundangjaesaeng hospital acted as a main reason. When considering the medical personnel stays in the hospital for a long time, it is likely to be develop as mass infection. In addition, there is non medical personnel such as patient's family, caregivers other than medical personnel such as doctor, nurse in the hospital, they often stay in the hospital without moving specially.

Therefore, if the outbreak of infectious disease in the medical facilities will bring the situation exposed to the possibility of mass infection, so it is necessary to minimize the staying time of non medical personnel in the hospital. Along with this, because analyzing their route is often impossible due to move within the building, so it is necessary to consider the plan to identify the chain of transmission of disease by utilizing the chain of infection between contact through the net work analysis methodology.

Fourth, the accurate information should be provided to nationals by establishing the realistic and reliable quarantine measure based on the empirical analysis result utilizing the data with the characteristics of region and business. For example, the convenience stores are frequently visited by many people as well as the confirmed patient but the staying time is within 2 minutes in average, which is significantly less than the 15 minutes suggested by WHO as a guideline.

In fact, no case of COVID-19 spread due to the contact in the convenience store was observed during the period of analyzing the study. In particular, the confirmed patient who is the member of 'Shincheonji' had worked for the convenience store for a long time but the additional confirmed patient was not discovered due to this. When considering this, the proper quarantine policy that fits the characteristics of visit place should be established and arranging the system for information analysis should be preceded vitally during this process. In addition, there was the case in which some local governments raised the anxiety of citizens by unclearly disclosing the name of visit place of the confirmed patient. Therefore, the effort to provide the accurate information on the visit place is needed.

V. Conclusion

Gyeonggi-do which secured various transportation means such as road, railway, port and airport can be characterized by the easy accessibility at home and abroad. Due to this, there is a risk that the spread of the disease can also occur rapidly, so it is an area with conditions in which the spread of infectious diseases can proceed explosively. The purpose of this study is to analyze the type and movement characteristics of the confirmed patients and to provide the political implications by collecting the route information of COVID-19 confirmed patients in Gyeonggi-do from January to April 2020. The routes identified from 237 confirmed patients collected from this study were 2,611 and 11 routes in average were identified from the confirmation until transferred to the hospital.

As a result of analysis based on the data established in this study, there occurred the case of infections that fall to all the type of WHO Guideline ① household and community, ② long-term living facilities, prisons, shelters, hostels, ③ healthcare setting, ④ transportation, ⑤ other (chapel, workplace, school, Private gathering, etc.). This study focused on ① household and community, ② healthcare setting, and ③ Other well-defined setting and gatherings (place of worship, workplace, schools, private social events) among these types. The following implications were derived from the three types.

First, the quarantine policy should be corrected by less focusing on long term shelter, prison type and the transportation type, which rarely appear in Korea. Second, the examples of 'other' items include many types, but the fact to reflect the situation in Korea well was identified through empirical analysis. Therefore, it is necessary to arrange the quarantine policy and related detail response instruction in which other types are main types. Third, identifying the characteristics of visit place of confirmed patients and corresponding properly can reduce social chaos.

The limits of this study are as follows. After COVID-19 infection level is upgraded to 'serious' stage, each local governments have managed the information of confirmed patients, the disclosure levels such as prevention of damage to the commercial district, the personal information of confirmed patients are different by the jurisdiction of each local governments, causing the serious deviation in the quality of regional data. The accuracy of the information disclosed by each local government was also greatly deviated. For example, the routes of some confirmed patients such as their house are managed by the community service center in the region which is estimated to be the residence of the confirmed patient, causing the error with the routes analyzed in this study.

In addition, there is a limit that can't identify entire Gyeonggi-do due to the absence of the information on the confirmed patient in Bucheon and Gwangmyeong because the data has been deleted since May 3, 2020. As such, the route analysis method of confirmed patients has inherent limitations in identifying the relationship between routes and spread in the building. Therefore, in the future, it is necessary to conduct in-depth research on the spreading route of the confirmed patients or super-spreader through the network analysis methodology between confirmed patients.

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